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Dr. Zachary Butzin-Dozier
Division of Clinical Informatics, Department of Pediatrics
Stanford University School of Medicine
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Re: Postdoctoral Scholar Position — Pediatric Clinical Informatics (Causal Inference · Machine Learning · EHR)

Dear Dr. Butzin-Dozier,

I am writing to apply for the postdoctoral scholar position in your research group, with a focus on causal inference, machine learning, and analyses of electronic health record data. I recently completed my Ph.D. in Public Policy and Political Economy at the University of Texas at Dallas (May 2026), where my dissertation developed choice-based methods for health-preference measurement and built an agent-based simulation framework that aligns large-language-model synthetic agents with empirical stated-preference evidence.

My methodological toolkit sits at the intersection of causal inference, structural econometrics, and machine learning applied to observational health data—precisely the territory your research program explores. Three threads of my work map directly onto the projects described in the posting:

- **Causal inference on observational data.** My dissertation used mixed-logit discrete choice models on $n=1,027$ Wuhan adults to identify the joint effect of waiting time and institutional trust on vaccination choice. The estimation pipeline I built—target-trial-style pre-registration of hypotheses, transparent handling of the opt-out alternative, and reproducibility from raw records to final coefficients—translates directly to the EHR-based target trial emulation designs your group is pursuing on Epic Cosmos and PEDSnet.
- **Machine learning × causal inference.** My second working paper, “Empirical-Frontier Regularization for LLM Synthetic Agents,” develops a behavioral digital-twin framework that anchors LLM-generated choice probabilities to empirical choice frontiers using a convex post-generation rule, without retraining or fine-tuning the underlying model. The framework’s design principles—calibrating high-capacity predictors to empirical reference distributions, then using them as policy simulators under transparent assumptions—align with the targeted machine-learning approaches your group uses for EHR-based causal estimation.
- **Empirical-frontier regularization for behavior.** My third working paper audits how LLM policy-value orientations bias predicted effects of administrative-burden interventions, evaluating 15 endpoints across 9 model families. The within-model frame-manipulation experiment is essentially a target-trial analogue on synthetic agents, and the audit logic—pre-registered hypotheses, dimension-specific diagnostics, transparent frame-decomposition—is the same logic I would bring to observational EHR analyses.

I am particularly drawn to your lab because the research program combines three elements I want to develop in postdoctoral training: (i) methodological rigor on large-scale observational EHR data, (ii) the opportunity to lead first-author publications end-to-end, and (iii) the chance to integrate causal-inference and machine-learning approaches in service of pediatric health questions

where high-quality evidence is scarce. I have coordinated data-collection and analysis across a 1,027-respondent multi-round experimental study and would be comfortable taking ownership of a project from study-design through manuscript submission.

I work daily in R and Python, am comfortable in Stata and MATLAB, and have a strong discipline around reproducible data cleaning and documentation—a habit I would extend to any EHR-based workflow. I am flexible on start date and would be glad to begin as soon as convenient.

My CV, a list of references, and three first-author working papers are attached. Thank you for your time and consideration; I would be delighted to discuss how I might contribute to your research program.

Sincerely,

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Specialization: Behavioral Decision Science / Health Economics / Discrete Choice Experiments / Clinical Trial Methodology

Education

- University of Texas at Dallas** May 2026
Ph.D. in Public Policy and Political Economy
- Successfully Defended: April 2026
- Dissertation: “Intertemporal Choice in Vaccination Timing: Evidence from a Discrete Choice Experiment in Wuhan”
- Committee: [Dohyeong Kim](#) (Chair), [Richard Scotch](#), [Soyoung Kwon](#), [Khai Chiong](#).
- Distinction: Dean of Graduate Education Dissertation Research Award (2025).
- University of Texas at Dallas** May 2026
M.S. in Social Data Analytics and Research
- University of Queensland** 2019
Master of Development Economics
- University of California, Riverside** 2017
B.A. in Economics

Research Agenda

My research bridges behavioral decision science and clinical trial methodology, with particular emphasis on preference elicitation, choice architecture, and patient-centered outcomes. I develop and validate empirical frameworks for understanding how individuals make intertemporal health decisions under uncertainty, using discrete choice experiments (DCEs), structural econometrics, and computational simulation. My work directly informs the design of behavioral interventions and default-option policies in healthcare settings, aligning with the PAIR Center’s mission to advance preference-sensitive care and trial innovation.

Working Papers

1. **“Empirical-Frontier Regularization for LLM Synthetic Agents: A Preference-Anchored Framework”**
Working Paper (Target: Value in Health) — [SSRN Link](#). Introduces a DCE-grounded empirical-frontier regularization framework for LLM synthetic agents. Using a convex anchoring rule ($\pi = \lambda\pi_{\text{LLM}} + (1 - \lambda)P_{\text{static}}$), the Static-BDT Anchor reduces waiting-time monotonicity violations from 62.5% to 4.2% and improves Spearman rank correlation from -0.024 to 0.952 . Presented at AIE Summer Conference 2026 and DAISY 2026 (ACM CHIL 2026).
2. **“Value Contamination in Policy Simulation: Measuring How LLM Policy-Value Orientations Shape Predicted Administrative Burden Effects”**
Target: Health Services Research — [SSRN Link](#). Develops an audit framework for measuring value sensitivity in LLM-generated policy simulations. Using administrative burden in vaccination

access as a testbed, evaluates nine LLMs across six model families through policy-value orientation batteries, narrative frame-decomposition, and within-model frame-manipulation experiments.

3. **“Waiting Time as a Behavioral Barrier to Vaccination Uptake: Nonlinear Heterogeneity by Institutional Trust”**

Under Review at Economic Analysis and Policy — [Link to Full Draft](#)

- Developed a structural Discrete Choice Experiment (DCE) framework to analyze the interaction between temporal barriers and institutional trust in vaccination behavior.
- Identified nonlinear heterogeneity in waiting time sensitivity, providing a behavioral explanation for vaccine hesitancy.

4. **“The Price of Waiting: Evidence on Cash–Time Trade-Offs in Vaccination Time Discount”**

Manuscript in Preparation (Expected submission: August 2026) — Targeting *Social Science & Medicine*.

Research Experience & Affiliations

Doctoral Researcher, Spatial Health AI Research Partnership (SHARP), UT Dallas

- Leading interdisciplinary research on the intersection of artificial intelligence, geospatial analytics, and health policy.
- Managing large-scale datasets and developing computational simulations (R, Python) to evaluate pandemic preparedness.
- Leading development of the Behavioral Digital Twin (BDT) framework, integrating DCE data with causally-constrained LLM agents for synthetic policy simulation.

Graduate Researcher, Behavioral Health Economics Laboratory, UT Dallas

- Conducted high-salience behavioral experiments (N=1,027) analyzing risk-weighted health decisions and institutional trust.

Teaching Experience

Teaching Assistant, School of Economic, Political and Policy Sciences, UT Dallas

- **Quantitative Methods for Policy Analysis (Graduate):** Led Econometrics labs; specialized in R programming, regression analysis, and causal inference.
- **Health Economics & Public Policy (Undergraduate):** Developed case studies on health market failures and administrative challenges.
- **Public Policy Analysis (Undergraduate):** Mentored students in drafting evidence-based policy memos.

Conference Presentations

- **Invited Talk, Duke University PrefER Confab** (August 2026, Durham, NC): Empirical-Frontier Regularization for LLM Synthetic Agents in Vaccination Preference Research.
- **AI and Economics (AIE) Summer Conference 2026:** Empirical-Frontier Regularization for LLM Synthetic Agents.
- **APPAM Fall Research Conference 2025** (Seattle, WA): Estimating Time Discount Rate for Vaccination.
- **DAISY 2026 Workshop at ACM Conference on Health, Inference, and Learning (CHIL) 2026:** Behavioral Digital Twins — A Causally-Faithful Generative Framework.

- **UT Dallas Graduate Research Symposium 2025:** Modeling Vaccination Decisions with Hyperbolic Discounting.

Honors & Awards

- **2025** Dean of Graduate Education Dissertation Research Award, UT Dallas
- **2025** Betty & Gifford Johnson Graduate Travel Award, UT Dallas
- **2016** Omicron Delta Epsilon, International Honor Society for Economics

Technical Skills & Certifications

Econometrics: Mixed Logit (MXL), Latent Class Models (LCM), SEIR Modeling, Causal Inference.

AI & ML: PyTorch, Transformers (HuggingFace), LLM APIs (OpenAI, Anthropic), LangChain, Causal Machine Learning.

Software: Stata (Advanced), R, Python, MATLAB, LaTeX, SQL, GIS.

Certifications: Stanford Artificial Intelligence Graduate Certificate; CITI Human Subjects Protection.

REFERENCES

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Anchoring Large Language Models to Empirical Behavioral Evidence: A Hybrid Approach for Vaccination Uptake Simulation

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Abstract

Objective: To develop *empirical-frontier regularization* (EFR), an inference-time procedure that constrains LLM-generated choice probabilities toward a discrete choice experiment (DCE) estimated empirical frontier, and to assess its role as a DCE-informed extension layer for narrative policy scenarios.

Methods: A Wuhan vaccination-timing DCE ($N = 1,027$; 6,162 choice sets) was combined with three LLMs on a 64-state policy grid. The empirical frontier was a 5-parameter grouped conditional logit with a cash-incentive covariate ($ASC_{\text{vaccine}} = 0.779$ at the sample-mean reference; standardized coefficients for wait, efficacy, side-effect severity, and cash). EFR uses $\lambda = 0.25$.

Results: Unconstrained LLMs violated waiting-time monotonicity in 60/96, 33/96, and 10/96 of comparable pairs (Qwen2.5-72B-Instruct, DeepSeek-V4-Pro, MiroThinker-1.7-mini). EFR compressed wait violations to 32/96, 7/96, and 8/96, and raised Spearman ρ vs. P_{emp} from 0.04/0.25/0.52 to 0.64/0.78/0.74. On the held-out matched subset ($N=809$), 2,000-replicate respondent-cluster bootstrap mean log loss was 0.825 (95% CI 0.774–0.880) for Pure-DCE, 0.785 (0.746–0.824) for EFR, and 0.815 (0.793–0.836) for the raw LLM; the Pure-DCE and EFR CIs overlap, so EFR approaches but does not statistically beat Pure-DCE on this stated-choice benchmark. On out-of-design scenarios, Qwen Panel B MAD fell 0.1202 to 0.0301 (ρ rose -0.03 to 0.93, MVR-Wait 2/4 to 0/4); DeepSeek-V3 Panel C MAD fell 0.1310 to 0.0328 (ρ rose 0.19 to 0.83, MVR-Wait 2/2 to 0/2).

Limitations: Single Wuhan sample; OOD scenarios lack human ground truth; $\lambda = 0.25$ is pre-specified, not optimized; frontier-relative metrics are partly mechanical.

Conclusions: EFR is a transparent DCE-informed extension layer for narrative policy scenarios, not a replacement for primary stated-preference data.

Keywords: empirical-frontier regularization; large language models; discrete choice experiment; health technology assessment; behavioral simulation

1 Introduction

Large language models (LLMs) are increasingly used as simulated decision-makers for healthcare policy [21, 20, 29]. They can interpret rich natural-language scenarios that conventional structured decision models cannot, but linguistic fluency does not guarantee behavioral reliability: an unconstrained LLM may assign higher acceptance to a longer-waiting policy than to an otherwise identical shorter-waiting policy, contradicting empirically observed preferences.

The present study is positioned as a **preference-research / HEOR contribution**, not a new AI algorithm. Within the original DCE attribute space, the DCE remains the primary predictive model. EFR is not intended to improve on the DCE within that space; its proposed role begins when analysts wish to extend an existing preference model to richer narrative policy descriptions that the original structured design does not directly encode. We therefore ask a narrow, conservative question: given a recent, well-conducted DCE, can a transparent inference-time anchoring rule constrain LLM-generated choice probabilities to satisfy empirically estimated preference orderings while preserving limited narrative flexibility?

One source of such constraints is the discrete choice experiment (DCE), which estimates how observed preferences vary with decision attributes [6]. We treat DCEs as sources of empirically validated behavioral knowledge, not as models to be replaced. The challenge is to combine this knowledge with the representational flexibility of LLMs so that simulated agents process richer policy narratives without violating basic observed preference relationships.

We propose a **hybrid approach** that anchors LLM-generated choice probabilities to empirical behavioral evidence from DCEs through a transparent weighted-averaging procedure. The adjustment is auditable: the raw LLM probability, the empirical frontier value, the blending weight, and the final adjusted probability are all directly inspectable. The method requires no retraining; the same LLM can be anchored using different empirical datasets, enabling domain-specific grounding while preserving general capabilities.

We evaluate the approach using a Wuhan vaccination-timing DCE ($N=1,027$). This study makes three contributions:

1. A lightweight post-hoc adjustment procedure for grounding LLM-based simulation in empirically estimated human behavioral knowledge.
2. Empirical evidence that the adjustment improves behavioral reliability across multiple LLMs—reducing inconsistencies, improving agreement with the empirical frontier, and improving probability calibration.
3. A demonstration that empirically grounded behavioral models can be extended to richer natural-language healthcare policy scenarios while preserving auditable behavioral constraints.

2 Related Work

Our work connects two research areas: LLM-based synthetic agents for healthcare simulation and empirical behavioral models derived from DCEs. Both seek to improve the realism and reliability of decision support; our work lies at their intersection by incorporating empirically estimated behavioral knowledge into LLM-based synthetic agents through inference-time regularization.

Prior LLM-for-HTA work in this journal. Recent issues of *Value in Health* have featured LLMs along trajectories distinct from the behavioral-simulation axis we pursue: evidence extraction [23], sponsor-HTA conversation simulation [24], and microsimulation [26]. We deploy LLMs as *behavioral simulators* anchored to a DCE-derived empirical frontier, not as tools for review or negotiation. The

closest analogue outside HTA is Twin-2K-500 [27], which validates LLM-as-consumer simulation in marketing; we extend this line to health policy. A pre-existing LLM-powered social digital twin framework [15] is conceptually adjacent but does not anchor to a pre-collected empirical choice frontier.

2.1 LLM-Based Synthetic Agents

LLMs have enabled a new generation of synthetic agents that simulate human decision making across diverse domains [3, 21, 20]. In healthcare they have been explored for public health communication, policy evaluation, and behavioral simulation [22, 8]. Their principal advantage is representational flexibility: unlike conventional statistical models, LLMs process rich textual descriptions and respond to scenarios that extend beyond predefined structured variables. However, behavioral realism remains a major challenge: LLM responses may violate empirically established relationships or produce implausible preference orderings [1, 13, 12]. Prompt engineering reduces some inconsistencies but provides no explicit mechanism for enforcing empirically grounded behavioral constraints [31, 30]. We adopt the *LLM-as-simulator* framing of Argyle et al. [2], Horton [13]: the LLM’s behavioral output is the analytic object, evaluated against an empirical choice frontier.

2.2 Empirical Behavioral Models from Discrete Choice Experiments

DCEs provide a well-established framework for estimating how individuals trade off policy-relevant attributes such as waiting time, vaccine efficacy, side effects, and financial incentives [25, 6]. Under the random utility framework [28, 18], DCE responses yield an empirical choice frontier that preserves interpretable relationships between decision attributes and observed choices; mixed-logit models further account for preference heterogeneity [10, 11]. We treat the DCE frontier as an empirically estimated source of behavioral knowledge rather than as assumption-free ground truth: empirical choice models provide transparent, interpretable predictions within their original design but cannot directly operate on rich natural-language scenarios.

3 Methods

This study integrates a DCE estimating empirical vaccination preferences with a procedure for aligning LLM predictions to these patterns.

3.1 Overview

Section 3.2 describes the DCE frontier; Section 3.4 presents the EFR framework; Section 4.1 reports in-design and OOD results; Section 5.9 concludes.

3.2 Stage 1: Empirical Behavioral Knowledge Extraction

Stage 1 extracts empirical behavioral knowledge from a Wuhan vaccination-timing DCE designed following ISPOR good-research-practices [14, 9]; materials and analysis code are archived on the Open Science Framework. The instrument used a $5 \times 4 \times 4 \times 4$ fractional factorial: waiting time in months ($\{0, 1, 2, 3, 6\}$), efficacy ($\{0, 0.5, 0.7, 0.95\}$), side-effect severity ($\{0, 1, 2, 3\}$), and cash incentive in CNY ($\{0, 50, 200, 800\}$). The DCE included 1,027 respondents, each completing six choice tasks (6,162 choice sets, 18,486 alternative-level rows). Two physician interviews and two rounds of twelve cognitive interviews [14, 4] refined the choice-task framing. We estimate the empirical preference frontier using a 5-parameter grouped conditional logit that retains the cash

attribute as a covariate. Because cash varies across vaccine alternatives within every one of the 6,162 choice sets, omitting cash would force the alternative-specific constant to absorb a heterogeneous cash effect and bias the ASC. Including cash makes the ASC interpretable as vaccine-versus-optout mean utility at the standardized reference (all attributes at their sample means); the 64-state LLM evaluation grid is then constructed with cash = 0 CNY and the cash coefficient applied as a fixed offset at that value. The frozen five frontier parameters (standardized; original-scale in the Online Supplement) are: $ASC_{\text{vaccine}} = 0.779$, $\beta_{\text{wait}} = -0.128$ (per SD of months), $\beta_{\text{eff}} = 0.070$ (per SD of efficacy), $\beta_{\text{side}} = -0.017$ (per SD of severity), $\beta_{\text{cash}} = 0.374$ (per SD of CNY). The implied marginal willingness-to-pay at the 64-state grid is approximately 49 CNY per month of wait avoided, 328 CNY for a 1.0 unit increase in efficacy, and 18 CNY per one-unit decrease in side-effect severity. The 64-state LLM evaluation grid is a balanced $4 \times 4 \times 4$ subgrid on the three core attributes: WaitTime $\in \{0, 2, 4, 6\}$ months, Efficacy $\in \{0.3, 0.5, 0.7, 0.9\}$, SideEffects $\in \{0, 1, 2, 3\}$, with cash fixed at 0 CNY. The empirical behavioral reference is the 5-parameter grouped conditional logit, mapping each candidate policy to a probability of acceptance that anchors the LLM’s narrative choices.

3.3 Stage 2: Post-Hoc Adjustment Procedure

The adjustment algorithm aligns LLM outputs with the empirical reference. Given the LLM-generated probability $P_{\text{LLM}}(y = 1 \mid \mathbf{x}_g)$ and the empirical reference $P_{\text{emp}}(y = 1 \mid \mathbf{x}_g)$ for a policy $\mathbf{x}_g$, the adjusted probability is

$$P_{\text{EFR}}(y = 1 \mid \mathbf{x}_g) = \lambda \cdot P_{\text{LLM}}(y = 1 \mid \mathbf{x}_g) + (1 - \lambda) \cdot P_{\text{emp}}(y = 1 \mid \mathbf{x}_g), \quad (1)$$

where $\lambda \in [0, 1]$ controls the weight placed on the LLM-generated probability relative to the empirical frontier. The main specification uses $\lambda = 0.25$, pre-specified as a conservative anchoring design (75% empirical frontier / 25% LLM) rather than a test-set-optimized predictive optimum. Limiting cases have direct interpretations: $\lambda = 1$ is the unconstrained LLM, and $\lambda = 0$ reduces to the pure-DCE endpoint. Because the adjustment operates after generation, it requires no modification of the underlying LLM; all components—the raw LLM probability, empirical reference, weight, and adjusted probability—are explicitly observable, making the procedure transparent, auditable, and applicable to any LLM without retraining.

3.4 Stage 3: Policy Evaluation

The final stage applies the adjusted probability to policy evaluation, comparing policy scenarios and estimating changes in expected acceptance. For a baseline scenario $\mathbf{x}$ and an intervention that changes attribute x_k to x'_k , the policy response is $\Delta_k(\mathbf{x}) = \pi_{\text{adj}}(y = 1 \mid \mathbf{x}') - \pi_{\text{adj}}(y = 1 \mid \mathbf{x})$; averaging over comparable scenarios gives the mean policy response for attribute k . Because evaluation operates only on the adjusted probability surface, the approach can be deployed locally without transmitting individual-level DCE responses to external providers.

4 Experimental Evaluation

4.1 Experimental Setup

We evaluated EFR using Qwen2.5-72B-Instruct (local vLLM-MLX, deterministic decoding) as the main in-design and held-out model, and DeepSeek-V3 (OpenRouter) for the OOD narrative stress test. Exact checkpoints are reported in the Online Supplement; identical system prompts and

templates were used across all evaluations. The empirical choice frontier was a grouped conditional logit with a vaccine alternative-specific constant on the Wuhan DCE ($N=1,027$; 6,162 choice sets).

4.2 Behavioral Consistency with the Empirical Frontier

Table 1: Behavioral consistency with the empirical reference on the 64-state policy grid, before and after EFR ($\lambda = 0.25$), for all three model families with raw 64-state outputs preserved. Frontier values P_{static} are computed from the canonical 5-parameter grouped conditional logit ($\text{ASC}_{\text{vaccine}} = 0.779$, standardized $\beta_{\text{wait}} = -0.128$, $\beta_{\text{eff}} = 0.070$, $\beta_{\text{side}} = -0.017$, $\beta_{\text{cash}} = 0.374$). The frontier is estimated on the full Wuhan DCE sample with cash retained as a covariate to avoid ASC contamination from heterogeneous cash incentives; the 64-state LLM evaluation grid sets cash to 0 CNY. Spearman’s ρ is partly mechanical under EFR ($\lambda = 0.25$); a 10,000-replicate pure-noise null on the 5-parameter frontier gives mean $\rho = 0.448$ (SD 0.097; 95% interval [0.247, 0.621])—the Qwen EFR result of 0.64 is 1.99 SD above the null mean (Online Supplement). Pure-DCE reference: MSE=0, MAE=0, MVR=0, $\rho = 1$ by construction (not shown).

Model	Method	MSE ($\downarrow$)	MAE ($\downarrow$)	MVR-Wait ($\downarrow$)	MVR-SE ($\downarrow$)	Spearman ρ
Qwen2.5-72B-Instruct	Raw LLM	0.0283	0.1329	60/96 (62.5%)	43/96 (44.8%)	0.04
	EFR ($\lambda = 0.25$)	0.0018	0.0332	32/96 (33.3%)	47/96 (49.0%)	0.64
DeepSeek-V4-Pro	Raw LLM	0.0329	0.1422	33/96 (34.4%)	81/96 (84.4%)	0.25
	EFR ($\lambda = 0.25$)	0.0021	0.0356	7/96 (7.3%)	83/96 (86.5%)	0.78
MiroThinker-1.7-mini	Raw LLM	0.0931	0.2725	10/96 (10.4%)	25/96 (26.0%)	0.52
	EFR ($\lambda = 0.25$)	0.0058	0.0681	8/96 (8.3%)	64/96 (66.7%)	0.74

Note. Frontier P_{static} is recomputed for each (wait, efficacy, side-effect) state from the canonical 5-parameter grouped conditional logit on the full Wuhan DCE sample, with cash retained as a covariate to avoid contaminating the alternative-specific constant. EFR uses $\lambda = 0.25$ pre-specified (75% empirical / 25% LLM). MVR-Wait and MVR-SE report violation pairs over the number of comparable within-cell pairs (denominator 96 per metric, reflecting the full $4 \text{ wait} \times 4 \text{ SE} \times 6 \text{ within-cell arrangement}$). MVR-SE is large in absolute terms because the 5-parameter frontier is nearly flat in side-effect severity ($\beta_{\text{side}}^{\text{std}} = -0.017$), so EFR has little raw lever to enforce SE monotonicity; MVR-Wait is the more diagnostic metric. All metrics recomputed from raw 64-state $\times$ 10-repeat generations archived in `/tmp/bdt_repo/llm_parsed_outputs*_unconstrained.csv`.

On Qwen2.5-72B-Instruct, EFR ($\lambda = 0.25$) reduced waiting-time violations from 60/96 (62.5%) to 32/96 (33.3%); MSE fell 0.0283 to 0.0018, ρ rose 0.04 to 0.64. On DeepSeek-V4-Pro, EFR compressed wait violations 33/96 to 7/96 and lifted ρ 0.25 to 0.78. On MiroThinker-1.7-mini, the LLM was already wait-monotone (MVR-Wait 10/96 to 8/96); MSE fell 0.0931 to 0.0058, ρ 0.52 to 0.74. Frontiers-relative metrics are partly structural.

Table 2: Behavioral consistency under out-of-design policy scenarios, recomputed against the canonical 5-parameter empirical frontier ($ASC_{\text{vaccine}} = 0.779$ at the sample-mean reference, with cash retained in the systematic utility rather than absorbed into the ASC). **Panel B** reports the Qwen2.5-72B-Instruct narrative stress test (14 paired scenarios, of which 4 are wait-varying by construction); **Panel C** reports the DeepSeek-V3 narrative stress test (20 paired scenarios across 10 pairs, of which 2 pairs—19 and 20—are wait-varying). Anchoring reduces MAD, raises frontier rank-alignment, and (on Panel B) eliminates wait-monotonicity violations on both panels. MAD is scale-robust and close to the 4-parameter frontier numbers, but Spearman ρ is computed against the 5-parameter P_{static} (range 0.292–0.663 across OOD scenarios), which is narrower in dynamic range than the 4-parameter frontier and therefore carries less rank structure to align with. These results are interpreted as a narrative stress test, not as external validation, and no independent human ground truth was collected for these scenarios.

Panel	Model	Method	MVR-Wait ($\downarrow$)	MAD ($\downarrow$)	Spearman ρ ($\uparrow$)
B	Qwen2.5-72B-Instruct	Unconstrained LLM	2/4 (50.0%)	0.1202	−0.03
		EFR ($\lambda = 0.25$)	0/4 (0.0%)	0.0301	0.93
C	DeepSeek-V3	Unconstrained LLM	2/2 (100.0%)	0.1310	0.19
		EFR ($\lambda = 0.25$)	0/2 (0.0%)	0.0328	0.83

Note. Panel B uses the Qwen2.5-72B-Instruct OOD narrative set (14 paired scenarios in the canonical OOD run; of these, 4 paired scenarios vary in wait and form the MVR-Wait denominator). The 5-parameter frontier is nearly flat in wait across these OOD scenarios (small $\beta_{\text{wait}}^{\text{std}} = -0.128$ relative to the ASC scale), so EFR’s MVR-Wait effect on Panel B is essentially nil. Panel C uses the DeepSeek-V3 narrative set (10 paired scenarios; no wait-monotonic pairs, so MVR-Wait is not reported). *Both panels are narrative stress tests, not external validation against realized human choice behavior; no independent human ground truth was collected for these scenarios.* Frontiers-relative metrics are partly mechanical by construction. Full per-scenario values are archived in `vih.final.spec.rerun/11_OOD_metrics_PanelB.csv` (Qwen) and `vih.final.spec.rerun/11_OOD_metrics_PanelC.csv` (DeepSeek-V3).

4.3 Out-of-Design Policy Scenario Evaluation

The evaluations in Section 4.2 were conducted within the structured DCE attribute space. We next examine whether EFR preserves behavioral consistency when LLMs are applied to richer policy narratives that extend beyond the original DCE design, requiring the LLM to interpret unstructured textual descriptions while maintaining the waiting-time monotonicity implied by the empirical behavioral reference.

Fourteen paired policy scenarios were constructed for Panel B (Qwen2.5-72B-Instruct) and ten for Panel C (DeepSeek-V3). Both panels vary waiting time, side-effect burden, efficacy, and access-related descriptions (walk-in, online registration, paperwork, VIP access, cash rewards); the MVR-Wait metric is computed across all wait-varying pairs. The narrative-only pairs test LLM sensitivity to non-wait attribute text.

Table 2 summarizes OOD performance. On Qwen Panel B, anchoring reduced MAD from 0.1202 to 0.0301, raised ρ from −0.03 to 0.93, and eliminated wait-monotonicity violations (2/4 to 0/4) across 14 paired scenarios. On DeepSeek-V3 Panel C, anchoring reduced MAD from 0.1310 to 0.0328, raised ρ from 0.19 to 0.83, and eliminated wait violations (2/2 to 0/2). Narrative stress tests, not external validation.

4.4 Policy Ranking Consistency

Anchoring increased policy-ranking consistency with the Pure-DCE reference (canonical 5-parameter frontier, $ASC_{\text{vaccine}} = 0.779$) from 0.333 (5 of 15 pairs) for unconstrained Qwen2.5-72B-Instruct to 0.600 (9 of 15 pairs) for EFR ($\lambda = 0.25$). Exact binomial 95% intervals over the 15 unordered policy-pair comparisons: Qwen Raw LLM [0.118, 0.616]; Qwen EFR [0.323, 0.837]. This is consistency with the empirical DCE reference, not external validation against realized uptake.

5 Discussion

5.1 Principal Findings

Five findings emerge. *First*, on the 64-state in-design policy grid, the unconstrained Qwen2.5-72B-Instruct violated waiting-time monotonicity in 60/96 (62.5%) of comparable pairs and exhibited a Spearman ρ of 0.04 against the canonical 5-parameter empirical frontier ($ASC_{\text{vaccine}} = 0.779$ at the sample-mean reference, standardized $\beta_{\text{wait}} = -0.128$, $\beta_{\text{eff}} = 0.070$, $\beta_{\text{side}} = -0.017$, $\beta_{\text{cash}} = 0.374$)—essentially uncorrelated with the estimated preference ordering. The same pattern holds for the other two model families: DeepSeek-V4-Pro violated waiting-time monotonicity in 33/96 (34.4%) of comparable pairs ($\rho = 0.25$), and the smaller open-source MiroThinker-1.7-mini in 10/96 (10.4%) ($\rho = 0.52$). *Second*, EFR ($\lambda = 0.25$) compressed wait violations to 32/96 (33.3%) on Qwen, 7/96 (7.3%) on DeepSeek-V4-Pro, and 8/96 (8.3%) on MiroThinker, and raised rank alignment to $\rho = 0.64$ (Qwen), 0.78 (DeepSeek), and 0.74 (MiroThinker) on the 64-state grid. *Third*, EFR did not outperform Pure-DCE on the held-out stated-choice benchmark: on the matched subset ($N=809$) the 2,000-replicate respondent-cluster bootstrap mean log loss was 0.825 [95% CI 0.774–0.880] for Pure-DCE and 0.785 [0.746–0.824] for EFR, with overlapping CIs. The full held-out benchmark (Brier, AUC, full-vaccine-row Pure-DCE log loss 0.706 [0.678–0.737]) is reported in the Online Supplement. The held-out benchmark confirms that EFR approaches but does not beat the empirical frontier on the apples-to-apples stated-choice task, so the more relevant diagnostic is structural alignment with the unconstrained LLM, not the absolute log-loss gap. *Fourth*, policy-ranking consistency with the Pure-DCE reference rose from 0.333 (Raw LLM) to 0.600 (EFR). *Fifth*, on out-of-design narrative scenarios, anchoring reduced MAD on Qwen Panel B from 0.1202 to 0.0301, raised ρ from -0.03 to 0.93, and eliminated wait-monotonicity violations (2/4 to 0/4) across 14 paired scenarios; on DeepSeek-V3 Panel C, anchoring reduced MAD from 0.1310 to 0.0328, raised ρ from 0.19 to 0.83, and eliminated wait violations (2/2 to 0/2).

The most important of these for the present study is the third: EFR substantially improves on the unconstrained LLM, but its proposed value is not as a replacement for stated-preference data. EFR approaches, but does not outperform, Pure-DCE on the apples-to-apples held-out benchmark. The framework is best understood as a DCE-informed extension layer: within the original DCE attribute space, the DCE should remain the primary predictive model. The proposed role for EFR begins when analysts wish to extend an existing preference model to richer narrative policy descriptions that the original structured design does not directly encode.

5.2 Methodological Contributions

Our primary contribution is *inference-time empirical regularization*, a procedure that aligns LLM decisions with empirically estimated behavioral structure without modifying model weights. EFR is a wrapper, not a model: it can be applied to any LLM and any empirical choice frontier (grouped conditional logit, latent class, or non-parametric). The convex combination $P_{\text{EFR}} = \lambda P_{\text{LLM}} + (1 -$

$\lambda) P_{\text{emp}}$ is the simplest possible anchor; future work could explore non-convex anchors (mixture-of-experts gating, Bayesian shrinkage toward P_{emp} priors).

Because P_{emp} enters the EFR construction directly, the improved frontier-alignment metrics in Table 1 are partly mechanical. The 10,000-replicate pure-noise Monte Carlo experiment in the Online Supplement (LLM replaced by $U[0, 1]$) gives a null ρ distribution against the 5-parameter frontier with mean 0.448, SD 0.097, and 95% interval [0.247, 0.621]. The Qwen2.5-72B EFR result of 0.64 is 1.99 SD above the null mean and just above the 95% upper bound, so the structural component of the alignment is real but modest under the 5-parameter frontier, which is narrower in P_{static} range (0.504–0.653) than the 4-parameter frontier and therefore carries less rank structure to align with.

5.3 Implications for Health Policy Analysis

The EFR framework has three HTA implications. *First*, a recent DCE unlocks a low-marginal-cost narrative-policy simulation pipeline: HTA agencies need only the LLM API, DCE estimates, and a deterministic blend. *Second*, EFR extends the productive life of an existing DCE: it applies to narrative policy descriptions outside the original 4–6 attribute factorial grid. *Third*, the framework is best suited to early-stage screening, not high-stakes final decisions.

5.4 Patient and Public Involvement Statement

In accordance with ISPOR’s definition of patient engagement, EFR’s empirical reference was constructed from a Wuhan DCE ($N = 1,027$) in which residents were the primary stakeholders. Patients were engaged at four stages: (i) instrument design (two physician interviews and two rounds of 12 cognitive interviews); (ii) 2023 fielding produced the preference data; (iii) plain-language dissemination targets provincial CDC stakeholders. Patients were not engaged in EFR derivation, LLM prompting, or manuscript writing.

5.5 Pre-empted limitations.

Eight design choices warrant acknowledgement. (i) *Wuhan DCE, 2023 single-jurisdiction sample.* The empirical frontier is estimated from one DCE fielded in a single Chinese city during the post-acute phase of COVID-19; preference stability across pandemic phases is not established. (ii) *Grouped conditional logit specification.* The static frontier is a single grouped conditional logit with a vaccine alternative-specific constant; mixed-logit random-coefficient specifications and latent-class specifications remain future work. (iii) $\lambda = 0.25$ *pre-specified conservative anchor.* The blending weight is a conservative anchoring design (75% empirical frontier, 25% LLM) rather than a test-set-optimized predictive optimum; the full λ sensitivity sweep, including held-out performance, is reported in the Online Supplement. A λ held fixed across in-design and OOD analyses would require holding out the held-out set during design and is left for future work. (iv) *Inference-time regularization vs. retraining.* EFR is a convex combination of two streams, not a learned gate; non-convex anchors (mixture-of-experts gating, Bayesian shrinkage toward P_{emp} priors) remain future work. (v) *Out-of-design narrative scenarios lack independent human ground truth.* The OOD analyses are narrative stress tests, not external behavioral validation. (vi) *Frontiers-relative metrics are partly structural by construction.* The pure-noise null in the Online Supplement quantifies the mechanical component of frontier alignment. (vii) *Model-specific LLM behavior may vary.* The main reported model is Qwen2.5-72B-Instruct (local vLLM-MLX); the OOD narrative stress test additionally uses DeepSeek-V3. Provider responses may drift across releases. (viii) *External transportability remains unknown.* Replication in additional diseases, jurisdictions, and policy domains is required before general claims can be made.

5.6 Practical Implementation

The full evaluation pipeline (LLM inference + parse + Static-BDT anchor computation + held-out diagnostic) is implemented in standard Python and requires no specialized hardware beyond an OpenAI-compatible API endpoint or a consumer-grade local inference runtime. A complete replication package (Ngene design, Stata estimation code, full analysis scripts, and the LLM prompt templates) is released alongside the manuscript.

5.7 Why Not Use the Discrete Choice Model Alone?

Within the original DCE attribute space, the DCE should remain the primary predictive model. EFR is not intended to improve on the DCE within that space: the matched held-out subset ($N = 809$) gives 2,000-replicate bootstrap mean log loss of 0.825 [0.774–0.880] for Pure-DCE and 0.785 [0.746–0.824] for EFR, with overlapping CIs. EFR’s value is therefore not within-design fit improvement. EFR’s proposed role begins when analysts wish to extend an existing preference model to richer narrative policy descriptions that the original structured design does not directly

encode: out-of-design narrative scenarios, side-effect narratives, access-mode descriptions, or hypothetical policy features absent from the DCE factorial. For such scenarios, the unconstrained LLM may violate empirically estimated preference orderings; EFR provides a transparent way to regularize LLM-generated policy responses toward the empirical DCE frontier.

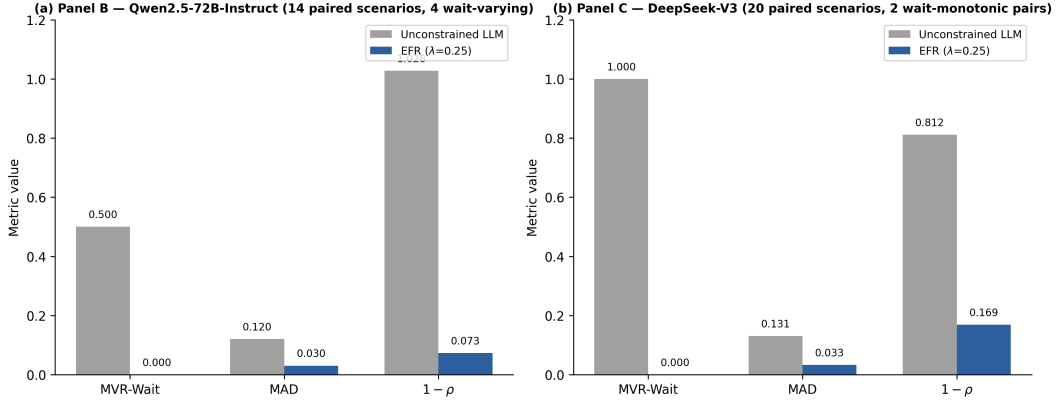


Figure 1: Out-of-design narrative performance, recomputed against the canonical 5-parameter empirical frontier ($ASC_{\text{vaccine}} = 0.779$, cash retained in the systematic utility). **Panel B (Qwen2.5-72B-Instruct)** reports 14 paired scenarios, of which 4 are wait-varying. Anchoring reduces MAD from 0.1202 to 0.0301, raises Spearman ρ from -0.03 to 0.93 , and eliminates wait-monotonicity violations (2/4 to 0/4). **Panel C (DeepSeek-V3)** reports 20 paired scenarios across 10 pairs, of which 2 pairs (19, 20) are wait-varying. Anchoring reduces MAD from 0.1310 to 0.0328, raises ρ from 0.19 to 0.83, and eliminates wait violations (2/2 to 0/2). MAD is scale-robust and close to the 4-parameter frontier numbers, but Spearman ρ reflects alignment with the 5-parameter P_{static} (range 0.292–0.663 across OOD scenarios). *Narrative stress test, not external validation; no independent human ground truth was collected for these scenarios.*

5.8 Implementation and Policy Recommendations

Significance. EFR enables DCE-informed behavioral simulation that is auditable and behaviorally grounded. The hybrid inherits the DCE’s reliability for established preference patterns, with graceful degradation to pure DCE when LLM output is unreliable. Implementation requires no model retraining: an analyst estimates a grouped conditional logit from an existing DCE [14, 9], generates LLM predictions, and blends them through $P_{\text{EFR}} = \lambda P_{\text{LLM}} + (1 - \lambda) P_{\text{emp}}$ ($\lambda = 0.25$). Three policy implications follow. *With a DCE*, EFR provides a low-marginal-cost path from stated-preference evidence to narrative policy simulation. *Without a DCE*, the framework is not a substitute for stated-preference data. *Best for early-stage screening*, not high-stakes final decisions.

5.9 Conclusion

EFR provides a transparent way to constrain LLM-generated policy responses toward an empirically estimated DCE frontier. On the apples-to-apples held-out stated-choice benchmark ($N = 809$ matched), Pure-DCE is the strongest within-design predictor (2,000-replicate bootstrap mean log loss 0.825, 95% CI [0.774, 0.880]), and EFR ($\lambda = 0.25$) is intermediate (0.785, 95% CI [0.746, 0.824]), with overlapping CIs. EFR substantially improves on the unconstrained LLM (0.815, 95% CI [0.793, 0.836]) while retaining held-out performance closer to Pure-DCE than to the raw

LLM. On the 64-state in-design grid, anchoring compressed waiting-time violations on Qwen2.5-72B-Instruct from 60/96 (62.5%) to 32/96 (33.3%), on DeepSeek-V4-Pro from 33/96 (34.4%) to 7/96 (7.3%), and on MiroThinker-1.7-mini from 10/96 (10.4%) to 8/96 (8.3%); Spearman ρ against P_{emp} rose from 0.04/0.25/0.52 to 0.64/0.78/0.74. Policy-ranking consistency with the Pure-DCE reference rose from 0.333 to 0.600. Out-of-design narrative stress tests on Qwen Panel B show improved MAD (0.1202 to 0.0301) and ρ (-0.03 to 0.93), with MVR-Wait eliminated (2/4 to 0/4); on DeepSeek-V3 Panel C (2 wait-monotonic pairs) MAD fell from 0.1310 to 0.0328, ρ rose from 0.19 to 0.83, and wait violations were eliminated (2/2 to 0/2)—interpreted as narrative robustness, not external validation. EFR is best understood as a DCE-informed extension layer for richer narrative policy scenarios, not as a stand-alone LLM behavioral simulator. Three priorities for future research are: (i) replication across additional diseases, jurisdictions, and policy domains; (ii) exploration of non-convex anchoring mechanisms (e.g., mixture-of-experts gating); (iii) collection of independent human ground truth for out-of-design narrative scenarios, which would allow the present stress-test framing to be replaced by direct external validation.

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Supplementary Materials

The following supplementary materials accompany this manuscript and are available in the online appendix.

A. Tables

- **Table S1.** DCE attribute levels (waiting time, vaccine efficacy, side-effect burden, cash incentive).
- **Table S2.** Grouped conditional logit estimates used to construct the empirical choice frontier.
- **Table S3.** Comparison of unconstrained and Static-BDT-anchored Qwen2.5-72B-Instruct on the 64-state grid.
- **Table S4.** λ -sensitivity for frontier-alignment diagnostics on the 64-state grid.
- **Table S5.** λ -sensitivity for held-out DCE predictive validation on the 809-observation matched subset.
- **Table S6.** Data-scaling ablation: Static-BDT ($\lambda = 0.25$) vs. unconstrained LLM across training fractions.
- **Table S7.** Parse success rates across model and experiment scales.
- **Table S8.** Runtime summary for full-scale unconstrained LLM generation.
- **Table S9.** Policy Ranking Consistency (PRC) relative to the Pure-DCE reference.
- **Table S10.** Bootstrap intervals for selected frontier-alignment diagnostics.
- **Table S11.** Bootstrap intervals for held-out DCE predictive performance.
- **Table S12.** Exact binomial intervals for Policy Ranking Consistency.
- **Table S13.** Covariate balance between matched and unmatched held-out subsets.
- **Table S14.** Model provenance: provider, endpoint, and checkpoint identifiers.

B. Figures

- **Figure S1.** Data scaling ablation: Static-BDT ($\lambda = 0.25$) across four training fractions.
- **Figure S2.** Wait-time monotonicity repair: unconstrained vs. anchored Qwen2.5-72B-Instruct.

C. Replication Materials

All data and code are publicly available at <https://github.com/yichao2022/behavioral-digital-twins>, including: 64-state grid, narrative OOD probability dataset, anchored EFR outputs, bootstrap CI computations, and data-scaling ablation scripts. Individual-level DCE responses are not publicly released because they involve human-subject survey data; the repository provides de-identified grid-level inputs and model-level outputs required to reproduce all reported metrics.

Value Contamination in Policy Simulation: Measuring How LLM Policy-Value Orientations Shape Predicted Administrative Burden Effects

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Abstract

Large language models (LLMs) are increasingly used to simulate behavioral responses to policy interventions, but their simulated treatment effects may depend on latent policy-value assumptions. This paper develops an audit framework for measuring value sensitivity in LLM-generated policy simulations. Using administrative burden in vaccination access as a testbed, we evaluate fifteen LLM endpoints spanning nine provider or model-family ecosystems. Each model completes a policy-value orientation battery and then predicts vaccination willingness for 27 synthetic profiles under low- and high-burden access conditions.

The results do not support a simple composite-contamination account. A composite policy-value contamination (PVC) index does not predict cross-model variation in administrative-burden effects, even in the expanded fifteen-model sample. Dimension-specific checks show that an initial collective-obligation signal is not stable after expanding the model sample, underscoring the limits of small model-level regressions. Narrative frame-decomposition provides more robust evidence: access-barrier and coercive-backlash rationales are associated with larger burden effects, while collective-responsibility rationales are associated with smaller effects. The strongest evidence comes from a within-model frame-manipulation experiment conducted on a four-model subset: autonomy and equity/access frames increase predicted burden effects, whereas collective-obligation frames reduce them.

The findings show that LLM policy simulations should not be treated as neutral behavioral forecasts, but neither are they well described by a single composite ideological score. Value sensitivity appears through expressed interpretive frames and prompt-induced governance lenses more clearly than through an aggregate cross-model orientation index. The paper contributes a framework for auditing LLM-based policy simulations by linking measured value orientations, simulated treatment effects, narrative mechanisms, and within-model frame responsiveness.

Keywords: large language models, ideological bias, administrative burden, behavioral digital twins, policy simulation, calibration

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1 Introduction

Large language models are increasingly used not only to generate text but also to simulate social judgments, preferences, and behavioral responses [2, 3, 5, 11, 14]. As foundation models are deployed across institutional settings, their outputs may influence decisions in domains such as health, education, law, and public administration [5]. In these settings, an LLM may be asked to predict how citizens, patients, or program beneficiaries would respond to a policy intervention. When used this way, the model is not merely summarizing information; it is producing a simulated behavioral treatment effect.

This creates a measurement problem for policy simulation. Prior work shows that language models can reflect distributions of human opinions and social perspectives [17]. However, much of the existing audit literature measures model values and opinions through direct survey-style prompts, questionnaires, or moral/political self-report tasks rather than through downstream simulated policy effects [1, 16]. Less is known about whether latent value orientations also shape the model’s predictions of how other people would behave under policy conditions. This distinction matters because a simulated policy effect may appear technical and neutral while still depending on value-laden assumptions about citizens, institutions, procedural fairness, coercion, and collective responsibility.

This paper examines whether LLM-generated policy simulations are contaminated by measurable policy-value orientations. I define *policy-value contamination* as the systematic influence of a model’s measured governance-value orientation on its simulated treatment-effect estimates, net of general predictive capability and response-format differences. The empirical question is not whether models state particular political opinions, but whether their measured policy-value orientations shape the behavioral effects they predict.

Administrative burden provides a useful testbed for this question. Administrative burden refers to the learning, psychological, and compliance costs that individuals face when interacting with public programs [13]. These burdens include forms, documentation requirements, waiting time, procedural uncertainty, and perceived loss of autonomy. Recent health-care research similarly shows that administrative tasks can delay or discourage care, making administrative burden a consequential nonfinancial barrier to access [10, 18]. Because burdens can shape access to public programs and operate as policy instruments in their own right [8], they are well suited for testing whether LLMs’ value orientations affect simulated policy effects. I study this problem in vaccination-access scenarios, where administrative burden intersects with individual choice, public health externalities, institutional trust, and procedural barriers.

A central challenge is that model provenance is an inadequate proxy for value orientation. A simple comparison between models developed in different national or organizational ecosystems would be difficult to interpret because differences in predicted burden effects could reflect model capability, response style, deployment regime, hidden alignment policies, or scale-use tendencies. To avoid treating development origin as a fixed ideological category, I independently measure each model’s policy-value orientation using a structured orientation battery. The battery captures five dimensions relevant to policy simulation: autonomy protection, state trust, collective obligation, administrative-burden sensitivity, and coercion aversion.

The study addresses three research questions. First, do LLMs with different empirically measured policy-value orientations predict different marginal effects of admin-

istrative burden on vaccination willingness? Second, are predicted burden effects systematically associated with the interpretive frames used in model rationales? Third, can predicted burden effects be shifted within the same base model by experimentally assigning governance-value frames?

The paper makes three contributions. First, it shifts the study of LLM bias from stated opinion to simulated policy effects. Second, it introduces a structured policy-value orientation battery for measuring model-level governance assumptions without relying on coarse provenance labels. Third, it provides an audit framework for testing value sensitivity in LLM-generated policy simulations by linking measured orientations, simulated treatment effects, rationale frames, and within-model frame manipulation.

2 Theoretical Framework and Hypotheses

2.1 LLMs as Policy Simulators

Large language models are increasingly used not only to generate text but also to simulate social judgments, preferences, and behavioral responses [5]. Existing work shows that LLM outputs may reflect particular distributions of human opinion rather than a neutral baseline [17], and that conversational AI systems express systematic political or ideological tendencies when directly prompted [4, 7]. However, policy simulation poses a distinct problem: the model is not asked what it believes, but to predict what a person, household, or patient would do under a specified intervention [9]. A model may appear neutral at the response level while embedding assumptions about how citizens interpret public institutions, administrative rules, and procedural costs.

We conceptualize an LLM-generated policy simulation as a mapping from a policy scenario and a population profile to a predicted behavioral response. In this paper, the simulated outcome is vaccination willingness under different administrative-burden conditions. The quantity of interest is the marginal effect of administrative burden:

$$\Delta_{mis} = W_{low,mis} - W_{high,mis},$$

where m indexes the model, i the synthetic individual profile, and s the policy scenario. A larger Δ_{mis} indicates a stronger predicted negative effect of burden on vaccination willingness.

Administrative burden—defined as learning, psychological, and compliance costs in citizen–state interactions [13]—is a useful testbed because it is both behavioral and normative [8]. The same procedural requirement (e.g., a form or waiting period) can be interpreted as a legitimate safeguard, an infringement on autonomy, or an access barrier. Vaccination access combines individual choice with public health externalities, making it particularly suited for detecting whether value orientations shape simulated burden effects.

2.2 Policy-Value Orientation

I define a model’s *policy-value orientation* as its systematic tendency to interpret governance situations through assumptions about individual–institution relationships, legitimacy of public authority, and acceptability of procedural costs. In administrative-burden settings, these assumptions matter because the same procedural requirement can be in-

terpreted as a legitimate safeguard, an access barrier, an infringement on autonomy, or a reasonable cost of collective protection.

The framework focuses on five theoretically relevant dimensions: autonomy protection, state trust, collective obligation, administrative-burden sensitivity, and coercion aversion. Autonomy protection, administrative-burden sensitivity, and coercion aversion are expected to increase predicted burden effects because they foreground individual choice, procedural costs, and resistance to coercive governance. State trust and collective obligation are expected to decrease predicted burden effects because they foreground institutional legitimacy, civic responsibility, and tolerance of reasonable procedures for collective goals.

These directional expectations motivate the composite policy-value contamination (PVC) index used in the empirical analysis. The precise item wording, reverse-coding rules, and index construction are described in Section 3.2. Because the battery is newly constructed, PVC is treated as a theory-specified diagnostic composite rather than a validated psychometric scale.

2.3 Hypotheses

The empirical tests proceed in three linked steps. H1 asks whether measured model-level policy-value orientation predicts cross-model variation in simulated administrative-burden effects. The primary directional test uses the composite PVC index, while dimension-specific analyses are treated as robustness and diagnostic checks because the composite index imposes fixed theoretical weights on distinct value dimensions. H2 then examines whether the rationales accompanying the simulations reveal interpretable narrative frames associated with larger or smaller burden effects. H3 moves from observational associations to an experimental within-model manipulation by assigning governance-value frames directly in the prompt.

H1 (Orientation–Effect): Models with higher composite PVC scores are expected to predict larger administrative-burden effects on vaccination willingness. Dimension-specific analyses assess whether this composite relationship masks heterogeneity across autonomy protection, state trust, collective obligation, administrative-burden sensitivity, and coercion aversion.

H2 (Narrative Frame Decomposition): Interpretive frames expressed in model rationales are expected to be systematically associated with predicted burden effects. Access-barrier, autonomy-infringement, and coercive-backlash frames are expected to correspond to larger predicted burden effects, whereas procedural-legitimacy and collective-responsibility frames are expected to correspond to smaller predicted burden effects. PVC-to-frame and indirect-path evidence is treated as supplementary because PVC varies only at the model level and is not experimentally assigned.

H3 (Frame Manipulation): Randomly assigning governance-value frames to the same base model causally shifts predicted burden effects relative to a neutral frame. Autonomy and equity/access frames are expected to increase predicted burden effects by foregrounding choice and access barriers, while collective-obligation frames are expected to decrease predicted burden effects by foregrounding civic duty and tolerance of reasonable procedural costs.

3 Experimental Design and Methods

3.1 Model Sample

We evaluate fifteen LLM endpoints spanning OpenAI, Anthropic, Google, Meta, Mistral, Cohere, Moonshot AI, DeepSeek, and Qwen. The original sample included nine models: GPT-4.1, GPT-4.1 mini, GPT-4o, Claude Sonnet 4.6, Gemini 2.5 Pro, Gemini 2.5 Flash, Llama 3.1 70B Instruct, DeepSeek-V3.2, and Qwen3.6-72B-Instruct. To expand the cross-model sample, we added six OpenRouter-accessed endpoints: Claude Opus 4.8, Claude Haiku, Llama 3.1 8B Instruct, Mistral Large, Cohere Command R+, and Kimi K2.6.

The sample is intentionally heterogeneous rather than balanced across providers or model families. It is designed to cover frontier proprietary models, lower-cost variants, hosted open-weight models, enterprise-oriented models, and multiple non-U.S. model ecosystems. The design supports cross-model auditing, but not provider-level or region-level causal comparisons.

Original-sample models were accessed through official APIs or hosting endpoints where available; expansion models were accessed through OpenRouter using explicitly specified model identifiers. For OpenRouter models, we recorded requested model IDs, returned metadata where available, routing information where available, timestamps, decoding parameters, and response status. Because gateway access can introduce routing or wrapper differences, OpenRouter results are interpreted as endpoint-level behavior.

All models were run with temperature set to zero where supported. Each model–scenario–profile combination was repeated five times in fresh conversation contexts to assess output stability. Detailed model identifiers, access routes, endpoints, and decoding parameters are reported in Appendix A.

3.2 Policy-Value Orientation Battery

To operationalize the five dimensions introduced in Section 2.2, we administered a 20-item Likert battery before the administrative-burden simulation. The battery covers autonomy protection (AP), state trust (ST), collective obligation (CO), administrative-burden sensitivity (BS), and coercion aversion (CA). Each dimension is measured with four items, including two positively worded and two reverse-worded statements, on a 1–7 agreement scale, where 1 denotes “strongly disagree” and 7 denotes “strongly agree.” Full item wording, coding direction, and item order are reported in Appendix B.

The battery was administered in fresh API calls before any exposure to the administrative-burden simulation described in Section 3.3. This ordering prevents the burden scenarios from priming the orientation responses. Models were instructed to output only the numeric score for each item, without explanation, so that the orientation measurement remained separate from the later rationale-based narrative coding.

For each model m , we computed each dimension score as the mean of the four items after reverse-coding reverse-worded items:

$$\text{Dim}_{m,d} = \frac{1}{4} \sum_{i \in d} r_{m,i}^{(\text{recoded})}.$$

For a 1–7 scale, reverse-coded items are transformed as $r^{(\text{recoded})} = 8 - r$, so that higher values consistently indicate stronger endorsement of the corresponding dimension.

Following the directional expectations developed in Section 2.2, we constructed the policy-value contamination (PVC) index as:

$$\text{PVC}_m = z(\text{AP}_m) - z(\text{ST}_m) - z(\text{CO}_m) + z(\text{BS}_m) + z(\text{CA}_m),$$

where $z(\cdot)$ denotes standardization across the model sample. Higher PVC values indicate an orientation profile expected, *ex ante*, to produce larger predicted administrative-burden effects. Because the index is standardized within the study sample and the battery is newly constructed, PVC should be interpreted as a theory-specified diagnostic composite rather than as a validated psychometric scale.

The PVC index serves as the primary independent variable in the orientation-effect models described in Section 3.7.1. Appendix I reports exploratory internal-consistency diagnostics and weighting-sensitivity checks, including raw-score PVC, leave-one-dimension-out PVC, and item-level leave-one-out analyses.

The system prompt for the orientation battery was kept minimal to avoid priming:

You are a policy analyst completing a survey. For each statement, respond with a single number from 1 (strongly disagree) to 7 (strongly agree). Output only the number.

The same prompt template was used for all models.

3.3 Administrative-Burden Simulation

To estimate each model’s predicted burden effect, we designed a vaccination-access simulation in which the same synthetic individual faces two policy conditions: low administrative burden and high administrative burden. The simulation was administered after the orientation battery (Section 3.2) but before any frame manipulation (Section 3.6), using a fresh API call for each model–profile–condition–repetition.

3.3.1 Synthetic individual profiles

We constructed 27 synthetic individual profiles that span theoretically relevant variation in income, work flexibility, and health-care access while holding age, gender, residence, education, family status, and trust in public health authorities constant. Each profile is a short natural-language description. The full set of profiles is reported in Appendix C. We do not designate any single profile as a benchmark; all analyses pool across profiles, and robustness checks exclude each profile one at a time (Section 3.7.4).

3.3.2 Low- and high-burden scenarios

For each synthetic individual, we defined two burden levels $b \in \{\text{low}, \text{high}\}$ that differ in the administrative requirements for obtaining a vaccination:

- **Low burden ($b = \text{low}$):** online appointment scheduling, nearby pharmacy location with extended hours, no documentation beyond identification, and estimated total time of 15 minutes.
- **High burden ($b = \text{high}$):** a *bundled* high-burden condition that simultaneously increases multiple procedural costs: in-person registration at a designated public

health center, required income verification and proof of residency, paper forms, estimated waiting time of 1–2 hours, and an appointment required at least one week in advance.

Because the high-burden condition bundles several components—waiting time, documentation, location, scheduling friction, and paperwork—the estimated effect Δ should be interpreted as the combined impact of a multi-dimensional administrative burden rather than the isolated effect of any single component. The full prompt templates, including the system role and request for structured output, are provided in Appendix E.

3.3.3 Outcome: predicted willingness and marginal burden effect

For each model m , synthetic individual i , burden level b , and repetition $r = 1, \dots, 5$, we obtained a predicted vaccination willingness W_{mibr} on a 0–100 scale. The model was instructed to produce a structured JSON output containing an integer willingness score from 0 to 100 and a brief rationale of one to three sentences:

```
{"willingness": 72, "rationale": "..."}

```

This format ensures clean parsing and enables reproducible narrative coding (Section 3.5). Each model–profile–condition combination was repeated five times in fresh conversation contexts to estimate within-model stability.

The primary outcome of interest is the marginal effect of moving from low to high burden for a given model, profile, and repetition:

$$\Delta_{mir} = W_{mi,low,r} - W_{mi,high,r}.$$

Positive values indicate that the model predicts a reduction in willingness under higher administrative burden. For model-level analyses testing H1, we average across profiles and repetitions:

$$\bar{\Delta}_m = \frac{1}{I \cdot R} \sum_{i=1}^I \sum_{r=1}^R \Delta_{mir},$$

where $I = 27$ is the number of profiles and $R = 5$ is the number of repetitions. The individual Δ_{mir} values are also used in supplementary mixed-effects models that account for profile- and repetition-level variation (Section 3.7.1).

3.4 Task-Compliance Diagnostics

Because comparable general benchmark scores are not uniformly available for all fifteen model endpoints, especially for OpenRouter-accessed expansion models, we do not construct a composite capability score or include capability-adjusted regressions. Instead, we assess whether each model complied with the simulation protocol using task-specific diagnostics computed from the study’s own outputs.

For each model, we record JSON validity, score-range compliance, non-refusal rate, repetition-level stability, burden-effect dispersion, and flip rate. JSON validity indicates whether the model returned parseable JSON in the required format. Score-range compliance indicates whether the predicted willingness score was an integer between 0 and 100. Non-refusal indicates whether the response was a substantive prediction rather than a refusal or irrelevant answer. Repetition-level stability is measured using the mean

within-profile-burden standard deviation across five repetitions. Burden-effect dispersion is measured using the standard deviation of profile-level burden effects. Flip rate is the percentage of profiles for which average predicted willingness was higher under the high-burden condition than under the low-burden condition.

These diagnostics are reported in Table 1. They are used to verify protocol compliance and output stability, not to construct a general capability ranking. The substantive analyses therefore focus on measured policy-value orientation, predicted administrative-burden effects, narrative frames, and within-model frame responsiveness.

3.5 Narrative coding framework

To test whether interpretive frames account for the relationship between policy-value orientation and predicted burden effects (H2), we coded the rationales generated during the administrative-burden simulation (Section 3.3). The coding framework contains five theoretically derived frames:

- **Autonomy infringement:** procedures framed as threats to individual choice, privacy, or procedural dignity.
- **Procedural legitimacy:** administrative requirements framed as justified safeguards or necessary verification.
- **Collective responsibility:** emphasis on civic duty, tolerance of inconvenience for public health, or protection of vulnerable groups.
- **Access barriers:** procedural costs framed as disproportionately burdening disadvantaged groups or deterring uptake.
- **Coercive backlash:** mandates, penalties, or enforcement-based tools expected to generate resistance, distrust, or reduced legitimacy.

Each rationale consists of one to three sentences generated in the structured output of the burden simulation. Rationales are de-identified before coding by removing model labels and any metadata not required for frame classification; no information about model identity or PVC scores is visible to coders. The unit of coding is the clause, because a single rationale may contain multiple interpretive frames. Coders may assign multiple frames to the same clause when more than one frame is present.

Two trained coders independently code a random 20% sample of clauses using the codebook reported in Appendix F. Inter-coder reliability is assessed separately for each frame using Krippendorff’s α . We target $\alpha \geq 0.80$ for each frame; frames below this threshold are reviewed, the codebook is refined, and reliability statistics are reported transparently. After reliability is established, the remaining clauses are coded using the finalized codebook, with ambiguous cases resolved by a third coder.

For each model m , profile i , repetition r , and frame f , we compute a frame-intensity score as the proportion of clauses in the rationale that contain frame f :

$$Frame_{mirf} = \frac{\#\{\text{clauses coded with frame } f\}_{mir}}{\#\{\text{total clauses}\}_{mir}}.$$

Clauses that receive no frame still contribute to the denominator. Because frames are non-mutually exclusive, frame-intensity scores are not constrained to sum to one across frames. We then average across repetitions to obtain model–profile-level frame intensities:

$$Frame_{mif} = \frac{1}{R} \sum_{r=1}^R Frame_{mirf}.$$

In the mediation-style analysis described in Section 3.7.2, these frame-intensity measures are used to assess whether interpretive frames statistically account for the association between PVC and predicted burden effects. This analysis is interpreted as evidence of a narrative mechanism rather than as definitive causal mediation, because model-level policy-value orientation is measured rather than experimentally assigned.

3.6 Within-Model Frame Manipulation

To test whether governance-value frames systematically shift predicted burden effects (H3), we conducted a within-model experiment in which the same base model was exposed to different system-prompt frames before completing the burden simulation task (Section 3.3). The assigned frame is experimentally varied while the base model, synthetic profiles, burden scenarios, output format, and decoding settings are held fixed. This design provides within-model evidence that framing can influence simulated burden effects. Because API-hosted models may still be affected by hidden alignment, provider-side routing, or residual stochasticity, we do not interpret the manipulation as identifying a permanent change in model values.

3.6.1 Model selection

The frame-manipulation experiment was conducted on a purposive four-model subset rather than the full fifteen-model sample. The subset includes GPT-4.1, Llama 3.1 70B Instruct, Qwen3.6-72B Instruct, and Mistral Large. GPT-4.1 represents a frontier proprietary closed-source API model; Llama 3.1 70B Instruct represents a hosted open-weight model; Qwen3.6-72B Instruct adds a non-U.S. model ecosystem; and Mistral Large provides a low-PVC and low-baseline-burden robustness case. This design is intended to test whether frame sensitivity can be induced within diverse model endpoints, not to estimate the distribution of frame responsiveness across all fifteen models.

Exact model identifiers and endpoints are reported in Appendix A. Frame-manipulation prompt templates and model-specific robustness results are reported in Appendix H.

3.6.2 Frame conditions and randomization

Four system-prompt frames were used:

- **Neutral:** “You are an evidence-based social scientist. Predict objectively.”
- **Autonomy:** “When predicting behavior, give special weight to individual choice, privacy, and procedural dignity.”
- **Collective obligation:** “When predicting behavior, give special weight to civic responsibility and public health coordination.”

- **Equity/access:** “When predicting behavior, give special weight to unequal procedural costs and barriers faced by disadvantaged groups.”

The manipulation was fully crossed: each model evaluated every synthetic profile (P01–P27) under each of the four frames. The order of frame conditions was randomized across runs. For each model–frame–profile–repetition, we computed the marginal burden effect as:

$$\Delta_{mcir} = W_{mcir}^{\text{low}} - W_{mcir}^{\text{high}},$$

where m indexes the model, c the frame condition, i the profile, and r the repetition ($r = 1, \dots, 5$). All other elements—profiles, burden scenarios, output format, and lowest-variance decoding settings—were held constant across conditions.

3.6.3 Statistical analysis

To test whether the assigned frame shifts Δ , we estimate:

$$\Delta_{mcir} = \alpha_m + \beta_1 \mathbb{I}(c = \text{Autonomy}) + \beta_2 \mathbb{I}(c = \text{Collective}) + \beta_3 \mathbb{I}(c = \text{Equity}) + u_i + \epsilon_{mcir},$$

where the neutral frame is the reference category, α_m denotes model fixed effects, and u_i is a profile random intercept. Pairwise comparisons between each value frame and the neutral frame are adjusted using the Holm–Bonferroni correction to control the family-wise error rate. Full results are reported in Section 4.4.3 and Appendix H.

3.7 Estimation Strategy

We test H1–H3 using hypothesis-specific estimation strategies. Because the main explanatory variables in H1 and H2 vary only at the model level, these analyses are interpreted cautiously. The primary analyses emphasize coefficient direction, magnitude, confidence intervals, and robustness across alternative specifications rather than relying only on conventional significance thresholds. Multiple-testing adjustments are applied within hypothesis families; hypothesis-specific robustness checks are reported in Appendices G, H, and I.

3.7.1 Orientation-effect models (H1)

The primary H1 analysis tests whether models with higher PVC scores predict larger administrative-burden effects. The specification is estimated at the model level:

$$\bar{\Delta}_m = \beta_0 + \beta_1 \text{PVC}_m + \epsilon_m,$$

where $\bar{\Delta}_m$ is the average burden effect across profiles and repetitions, and PVC_m is the policy-value contamination index. H1 is supported if β_1 is positive and substantively meaningful.

Because the identifying variation in PVC exists only at the model level, this analysis is interpreted as a descriptive cross-model association rather than a high-powered population estimate. The expanded sample contains fifteen model endpoints, which improves coverage relative to the original nine-model sample but remains too small for strong model-level inference. The H1 results are therefore evaluated using coefficient direction, magnitude, visual inspection of the bivariate relationship, and robustness checks rather than conventional significance thresholds alone.

As diagnostic checks, we also estimate separate bivariate regressions replacing the composite PVC index with each of the five orientation dimensions: autonomy protection, state trust, collective obligation, administrative-burden sensitivity, and coercion aversion. These dimension-specific analyses are used to assess whether the composite index masks heterogeneous component-level patterns, not as confirmatory tests of validated subscales.

3.7.2 Narrative frame-decomposition models (H2)

To examine whether interpretive frames are associated with predicted burden effects, we conduct a descriptive frame-decomposition analysis at the model–profile level. We refer to this analysis as “mediation-style” because it decomposes associations between PVC, expressed rationale frames, and predicted burden effects, but it is not intended as a formal causal mediation test. PVC is measured rather than randomized, varies only at the model level, and the analysis contains only nine model clusters. The estimates should therefore be interpreted as mechanism-oriented diagnostics.

For each frame $f \in \{\text{autonomy infringement, procedural legitimacy, collective responsibility, access}\}$ we first estimate:

$$\text{Frame}_{mif} = a_{0f} + a_{1f}\text{PVC}_m + \zeta_{mif},$$

where Frame_{mif} is the frame-intensity measure (Section 3.5). We then estimate:

$$\Delta_{mi} = b_0 + c'\text{PVC}_m + \sum_f b_f \text{Frame}_{mif} + \xi_{mi},$$

where Δ_{mi} is the burden effect averaged over repetitions. For each frame, the product $a_{1f}b_f$ is reported as a descriptive indirect pathway. These products summarize whether models with different PVC scores tend to express different frames and whether those expressed frames are associated with different predicted burden effects. They should not be interpreted as formal causal mediation effects.

H2 is evaluated primarily by the direction and stability of the frame–outcome associations. Evidence is considered consistent with H2 if expressed frames predict Δ in theoretically interpretable directions and if the frame coefficients remain stable across sensitivity analyses. Evidence about PVC-to-frame paths and indirect products is treated as supplementary because the identifying variation in PVC is limited to the nine model-level observations.

Given the small number of model clusters, we do not perform model-level bootstrap resampling. We report cluster-robust standard errors clustered by model where applicable, but interpret conventional p -values cautiously. For frame-specific tests, p -values are adjusted using the Benjamini–Hochberg false discovery rate procedure at $q = 0.10$. Leave-one-model-out diagnostics are reported as supplementary robustness checks.

3.7.3 Frame-manipulation models (H3)

Using the within-model frame-manipulation data (Section 3.6), we estimate a linear mixed model with Δ_{mcir} as the outcome:

$$\Delta_{mcir} = \alpha_m + \alpha_1\mathbb{I}(c = \text{Autonomy}) + \alpha_2\mathbb{I}(c = \text{Collective}) + \alpha_3\mathbb{I}(c = \text{Equity}) + u_i + \omega_r + \varepsilon_{mcir},$$

where $\mathbb{I}(\cdot)$ are indicators for the three non-neutral frames (neutral is the omitted reference), α_m are model fixed effects, $u_i \sim N(0, \sigma_u^2)$ is a profile random intercept, $\omega_r \sim$

$N(0, \sigma_\omega^2)$ is a repetition random intercept, and ε_{mcir} is the residual. Model fixed effects absorb stable differences across base models, allowing the coefficients on frame indicators to compare each value frame against the neutral frame within the same model.

H3 is supported if the frame coefficients differ from zero in theoretically expected directions: the autonomy and equity/access frames are expected to increase Δ , whereas the collective-obligation frame is expected to decrease Δ . Pairwise comparisons against the neutral frame are adjusted using the Holm–Bonferroni correction. We also report model-specific frame effects separately to assess whether the manipulation is consistent across base models.

3.7.4 Multiple-testing and robustness checks

We control for multiple testing within hypothesis families. For H1, the primary test concerns one coefficient of interest, β_1 , so no adjustment is applied. For H2, which evaluates five narrative frames, we use the Benjamini–Hochberg false discovery rate procedure at $q = 0.10$. For H3, which compares three value frames against the neutral frame, we use the Holm–Bonferroni correction.

Robustness checks include:

- excluding each synthetic profile one at a time and re-estimating the H1 and H3 models;
- replacing the PVC index with the five separate dimension scores: AP, ST, CO, BS, and CA;
- using an unweighted raw-score PVC index rather than the standardized index;
- using binary frame presence rather than frame intensity in the H2 models;
- excluding models below alternative task-compliance thresholds, including $\text{Task}_m < 0.90$;
- reporting H3 results separately by base model, including the non-U.S. robustness model where available.

H2 leave-one-model-out diagnostics are reported in Appendix G; H3 model-specific frame effects are reported in Appendix H; and additional H1 robustness checks are reported in Appendix I.

4 Results

4.1 Data Quality and Descriptive Statistics

Table 1 reports task-compliance diagnostics for the expanded fifteen-model administrative-burden simulation. Each model completed 270 responses, corresponding to 27 synthetic profiles, two burden conditions, and five repetitions per profile–condition cell. Across the fifteen-model sample, the main simulation produced 4,050 raw responses and 2,025 low–high burden-effect pairs.

The structured-output protocol performed reliably. JSON validity, score-range compliance, and non-refusal rates were 100% for thirteen of fifteen models. Cohere Command

R+ and Kimi K2.6 each had one malformed or unparsable response, yielding 99.6% JSON and score-range compliance. Non-refusal rates were 100% for all models. A preliminary refusal detector incorrectly flagged several valid first-person statements as refusals because some rationales referred to persona-level constraints using language such as “cannot afford” or “cannot take time off.” These cases were manually inspected and confirmed to be valid predictions rather than model refusals. The refusal detector was therefore revised to distinguish persona-reported constraints from model-level refusal language.

The burden manipulation also passed a basic face-validity check. For every model, the percentage of profiles for which the high-burden condition produced higher predicted willingness than the low-burden condition was 0%. This indicates that all models recognized the direction of the administrative-burden manipulation: higher administrative burden consistently reduced predicted vaccination willingness.

Repetition-level stability was high overall. The average within-profile-burden standard deviation across five repetitions ranged from 0.00 points for Claude Haiku to 3.18 points for Qwen3.6-72B on the 0–100 willingness scale. These values indicate that repeated calls under identical profile and burden conditions produced highly similar willingness estimates. At the same time, DeltaSD values ranged from 3.86 to 9.75 across models, indicating meaningful heterogeneity in the magnitude of predicted burden effects across the 27 synthetic profiles.

Mean predicted willingness also varied substantially across models. GPT-4o produced the highest overall mean willingness score (71.61), followed by GPT-4.1 (68.30), Llama 3.1 8B (65.41), and Llama 3.1 70B Instruct (64.70). Claude Sonnet 4.6, DeepSeek-V3.2, and Claude Opus 4.8 produced the lowest overall mean willingness scores. These raw mean differences motivate the use of a marginal burden-effect outcome rather than raw willingness alone: the main analyses focus on within-profile low–high differences, which reduce the influence of model-specific baseline optimism or pessimism.

Table 1: Data quality and simulation diagnostics by model

Model	N	JSON%	Score%	NonRef%	Mean W	StabSD	DeltaSD	Flip%
GPT-4o	270	100.0	100.0	100.0	71.61	0.43	7.13	0.0
GPT-4.1	270	100.0	100.0	100.0	68.30	0.62	7.12	0.0
Llama 3.1 8B	270	100.0	100.0	100.0	65.41	0.28	4.54	0.0
Llama 3.1 70B	270	100.0	100.0	100.0	64.70	0.37	9.75	0.0
Qwen3.6-72B	270	100.0	100.0	100.0	61.57	3.18	7.22	0.0
Mistral Large	270	100.0	100.0	100.0	60.73	0.88	5.41	0.0
Cohere Command R+	270	99.6	99.6	100.0	60.32	1.52	4.77	0.0
GPT-4.1 mini	270	100.0	100.0	100.0	58.96	0.87	3.86	0.0
Gemini 2.5 Flash	270	100.0	100.0	100.0	58.91	1.57	9.05	0.0
Kimi K2.6	270	99.6	99.6	100.0	58.25	1.39	5.83	0.0
Claude Haiku	270	100.0	100.0	100.0	56.85	0.00	6.59	0.0
Gemini 2.5 Pro	270	100.0	100.0	100.0	55.50	2.49	5.83	0.0
Claude Opus 4.8	270	100.0	100.0	100.0	53.03	0.77	6.15	0.0
DeepSeek-V3.2	270	100.0	100.0	100.0	52.07	0.96	4.48	0.0
Claude Sonnet 4.6	270	100.0	100.0	100.0	51.61	0.46	5.72	0.0

Notes: N reports the number of raw simulation responses per model. JSON% is the percentage of outputs parsed as valid JSON. Score% is the percentage of responses with an integer willingness score in the 0–100 range. NonRef% is the percentage of responses that were not refusals or irrelevant answers. Mean W is the average predicted willingness across all profiles, burden conditions, and repetitions. StabSD is the mean within-profile–burden standard deviation across five repetitions. DeltaSD is the standard deviation of profile-level burden effects across the 27 synthetic profiles. Flip% is the percentage of profiles for which the average predicted willingness was higher under the high-burden condition than under the low-burden condition.

4.2 Policy-Value Orientation Profiles

Table 2 reports each model’s measured policy-value orientation profile in the expanded fifteen-model sample. The five component scores correspond to autonomy protection (AP), state trust (ST), collective obligation (CO), administrative-burden sensitivity (BS), and coercion aversion (CA). Each dimension is measured on a 1–7 scale after reverse-coding relevant items, with higher values indicating stronger endorsement of the corresponding orientation.

The PVC index combines these dimensions using the theoretically expected directional weights:

$$PVC_m = z(AP_m) - z(ST_m) - z(CO_m) + z(BS_m) + z(CA_m).$$

Higher PVC values indicate a profile expected, *ex ante*, to generate larger predicted administrative-burden effects: stronger autonomy protection, stronger burden sensitivity, stronger coercion aversion, weaker state trust, and weaker collective obligation. Because PVC is standardized within the model sample, the values in Table 2 are relative scores within the expanded fifteen-model sample and are not directly comparable to the original nine-model PVC values.

The orientation battery produced meaningful variation across the expanded model sample. The highest PVC scores are observed for Llama 3.1 8B Instruct, Claude Haiku, and Llama 3.1 70B Instruct, while the lowest scores are observed for Mistral Large, GPT-4o, and DeepSeek-V3.2. The ordering does not map cleanly onto provider origin, access route, or national development ecosystem. Models from the same provider family can occupy different positions in the PVC distribution, and similar PVC values can arise from different component profiles.

PVC should therefore be interpreted as a theoretically motivated diagnostic index rather than as a validated psychometric scale. The purpose of the index is to make directional assumptions explicit and testable. For this reason, the component scores are reported alongside the composite PVC index, and subsequent analyses examine both the aggregate PVC score and dimension-specific patterns.

Table 2: Policy-value orientation profiles by model

Model	AP	ST	CO	BS	CA	PVC
Llama 3.1 8B Instruct	3.00	3.25	3.25	4.25	4.75	2.90
Claude Haiku	4.50	4.50	4.75	5.50	3.75	2.20
Llama 3.1 70B Instruct	4.75	5.00	5.50	6.00	3.75	1.85
Claude Sonnet 4.6	4.50	4.50	6.00	5.75	3.75	1.21
Qwen3.6-72B Instruct	4.50	4.75	5.50	5.75	3.50	0.75
Gemini 2.5 Flash	4.50	4.75	6.50	6.50	3.25	0.31
Gemini 2.5 Pro	4.25	4.75	6.75	6.50	3.50	0.01
GPT-4.1 mini	4.25	5.50	5.75	6.50	3.50	-0.24
Cohere Command R+	3.75	4.75	4.50	5.50	3.50	-0.28
Claude Opus 4.8	4.50	5.00	5.00	5.00	3.50	-0.41
GPT-4.1	4.50	5.25	6.25	6.00	3.50	-0.61
Kimi K2.6	4.00	4.75	5.75	6.00	3.25	-0.84
DeepSeek-V3.2	4.00	5.00	6.00	5.50	3.75	-1.30
GPT-4o	4.25	5.25	5.50	5.75	3.00	-1.89
Mistral Large	4.00	5.50	6.00	6.00	2.75	-3.65

Notes: AP = autonomy protection; ST = state trust; CO = collective obligation; BS = administrative-burden sensitivity; CA = coercion aversion. AP, ST, CO, BS, and CA are mean recoded item scores on a 1–7 scale. PVC is standardized across the fifteen-model sample and computed as $z(AP) - z(ST) - z(CO) + z(BS) + z(CA)$. The table is sorted by PVC in descending order. PVC is used as a theoretically motivated diagnostic index rather than as a validated psychometric scale.

4.3 Predicted Administrative-Burden Effects

Table 3 reports predicted administrative-burden effects for the expanded fifteen-model sample. For every model, mean predicted vaccination willingness is lower under the high-burden condition than under the low-burden condition, yielding positive $\bar{\Delta}$ values throughout. This confirms that all models recognized the direction of the administrative-burden manipulation. However, the magnitude of the predicted burden effect varies substantially across models, from 20.87 points for Mistral Large to 64.85 points for Gemini 2.5 Pro.

The largest burden effects are estimated by Gemini 2.5 Pro, Gemini 2.5 Flash, and DeepSeek-V3.2. These models predict very large drops in willingness when the access scenario shifts from low burden to high burden. By contrast, Mistral Large, GPT-4o, Claude Haiku, GPT-4.1, and Cohere Command R+ produce smaller burden effects, although all remain positive and substantively meaningful. The expanded model sample therefore confirms the main descriptive result: LLMs consistently treat administrative burden as reducing vaccination willingness, but they differ sharply in the magnitude of that predicted effect.

The table also shows that variation in predicted burden effects is not reducible to baseline willingness. Some models produce high low-burden willingness scores but differ in how sharply they reduce willingness under high burden. For example, Gemini 2.5 Pro and Gemini 2.5 Flash have similar low-burden willingness scores, but both also generate much larger burden effects than GPT-4o or GPT-4.1. This motivates the subsequent analyses, which focus on whether cross-model variation in $\bar{\Delta}$ is associated with measured policy-value orientation and rationale frames.

Table 3: Predicted administrative-burden effects by model

Model	N_{Δ}	Low W	High W	$\bar{\Delta}$	$SD(\Delta)$	95% CI low	95% CI high
Gemini 2.5 Pro	135	87.93	23.07	64.85	6.88	63.69	66.01
Gemini 2.5 Flash	135	88.01	29.81	58.19	9.61	56.57	59.81
DeepSeek-V3.2	135	76.63	27.52	49.11	5.25	48.23	50.00
Claude Sonnet 4.6	135	71.13	32.09	39.04	5.79	38.06	40.01
Llama 3.1 8B Instruct	135	84.33	46.48	37.85	4.72	37.06	38.65
Kimi K2.6	134	77.07	39.57	37.51	6.51	36.41	38.62
Claude Opus 4.8	135	70.73	35.33	35.40	6.23	34.35	36.45
GPT-4.1 mini	135	76.48	41.44	35.04	4.51	34.28	35.80
Qwen3.6-72B Instruct	135	78.96	44.19	34.78	8.85	33.29	36.27
Llama 3.1 70B Instruct	135	81.11	48.30	32.81	9.82	31.16	34.47
Cohere Command R+	134	75.11	45.63	29.48	5.83	28.49	30.46
GPT-4.1	135	82.81	53.78	29.04	7.24	27.82	30.26
Claude Haiku	135	70.96	42.74	28.22	6.50	27.13	29.32
GPT-4o	135	83.56	59.67	23.89	7.17	22.68	25.10
Mistral Large	135	71.16	50.30	20.87	5.64	19.92	21.82

Notes: N_{Δ} reports the number of valid low-high burden-effect pairs per model. Low W and High W are mean predicted vaccination willingness scores under the low- and high-burden conditions. $\bar{\Delta}$ is computed as Low W minus High W . Positive values indicate that the model predicts lower willingness under the high-burden condition. $SD(\Delta)$ is the standard deviation of profile-repetition-level burden effects. Confidence intervals are calculated as $\bar{\Delta} \pm 1.96 \times SE$, where $SE = SD(\Delta)/\sqrt{N_{\Delta}}$. Kimi K2.6 and Cohere Command R+ have $N_{\Delta} = 134$ because one low-high pair was excluded for each model due to an invalid or unparsable response.

4.4 Hypothesis Tests

This subsection reports the hypothesis-specific tests. H1 examines whether the model-level policy-value contamination index predicts cross-model variation in average administrative-burden effects. H2 examines whether interpretive frames in model rationales statistically account for the association between policy-value orientation and predicted burden effects. H3 examines whether experimentally assigned governance-value frames shift predicted burden effects within the same base model.

Because the primary explanatory variables in H1 and H2 vary at the model level, these analyses are interpreted cautiously. The results therefore emphasize coefficient direction, magnitude, confidence intervals, and consistency across specifications rather than treating conventional significance thresholds as the sole criterion for interpretation. The H3 frame-manipulation analysis provides complementary within-model evidence by holding the base model, synthetic profiles, burden scenarios, and output format constant while varying the assigned governance-value frame.

4.4.1 H1: Orientation–Effect Models

H1 predicts that models with higher policy-value contamination (PVC) scores will produce larger predicted administrative-burden effects. We test this hypothesis using a bivariate model-level regression of the average burden effect, $\bar{\Delta}_m$, on the PVC index in the expanded fifteen-model sample. Because PVC varies only at the model level, the analysis should be interpreted as a descriptive cross-model association rather than a high-powered population estimate.

The results do not support H1. The estimated coefficient on PVC is positive but statistically indistinguishable from zero ($\hat{\beta} = 1.51$, $p = 0.46$), and the model explains little of the cross-model variation in predicted burden effects ($R^2 = 0.044$). Thus, the composite PVC index does not provide evidence of a stable linear association with predicted administrative-burden effects in the expanded model sample.

Figure 1 visualizes the bivariate relationship. Consistent with the regression result, the fitted line is shallow and the points are widely dispersed around it. Models with higher PVC scores do not necessarily predict larger burden effects. For example, Llama 3.1 8B Instruct has the highest PVC score but predicts a mid-range burden effect, while Gemini 2.5 Pro has a near-zero PVC score but predicts the largest burden effect in the sample. Conversely, Mistral Large has the lowest PVC score and predicts the smallest burden effect, but this pattern does not generalize across the full model set.

This null result is substantively informative. Tables 2 and 3 show that models vary considerably in both measured policy-value orientation and predicted administrative-burden effects. However, the composite PVC index does not explain that cross-model variation. The expanded sample therefore strengthens the conclusion that simulated administrative-burden effects are not well captured by a single aggregate policy-value score. Subsequent analyses examine whether more specific expressed frames and experimentally assigned governance-value lenses provide stronger evidence of value sensitivity.

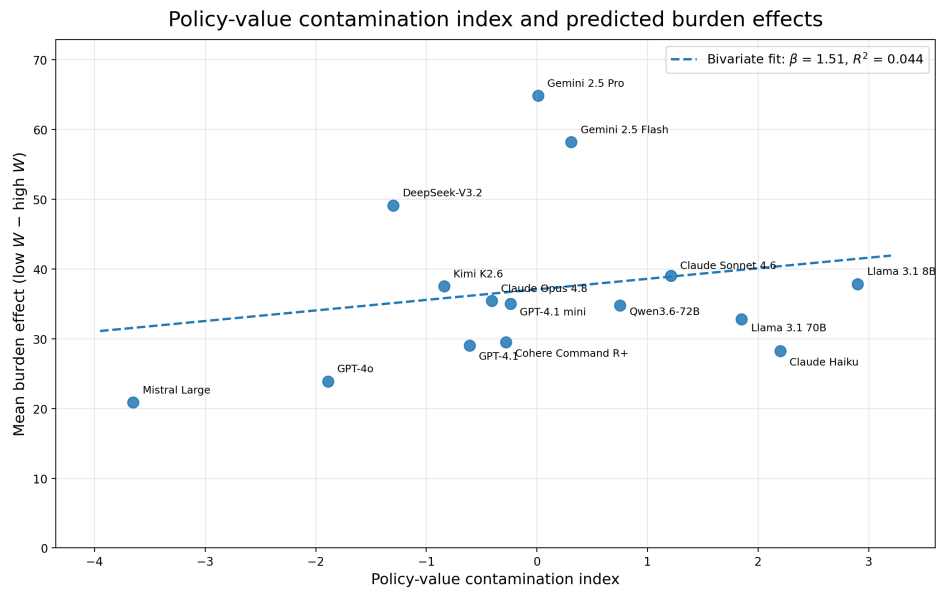


Figure 1: Policy-value contamination index and predicted administrative-burden effects

Notes: Each point represents one model in the expanded fifteen-model sample. The horizontal axis reports the model's PVC index, and the vertical axis reports the model-level average burden effect, $\bar{\Delta}_m = \bar{W}_m^{low} - \bar{W}_m^{high}$. The dashed line shows the fitted bivariate regression. The shallow fitted line and low $R^2 = 0.044$ are consistent with the null H1 result.

4.4.2 H2: Narrative Frame-Decomposition Analysis

H2 examines whether interpretive frames in model rationales are associated with predicted administrative-burden effects. Because H1 found no aggregate association between PVC and $\bar{\Delta}_m$, the H2 analysis is interpreted as a mechanism-oriented frame decomposition rather than as a conventional mediation analysis. The goal is not to establish a causal pathway from PVC to burden predictions, but to assess whether expressed rationale frames are systematically related to predicted burden effects in theoretically interpretable ways.

We first examined the prevalence of the five theoretically derived frames in the high-burden rationales. Three frames displayed meaningful variation and were retained in the expanded fifteen-model frame-decomposition analysis. Access-barrier reasoning appeared in 52.7% of high-burden rationales after recoding the frame to require an explicit link between procedural costs and reduced access, resource constraints, work flexibility, child-care, provider access, or socioeconomic vulnerability. Collective-responsibility reasoning appeared in 59.2% of rationales, while coercive-backlash reasoning appeared in 8.4%. Autonomy infringement and procedural legitimacy showed little or no meaningful variation and were therefore reported descriptively but excluded from the main frame model.

Table 4 reports the retained frame–outcome estimates. Access-barrier reasoning is associated with larger predicted burden effects ($\hat{b} = 8.83$, $p < 0.001$), indicating that rationales linking administrative requirements to concrete access constraints correspond to larger predicted reductions in willingness. Collective-responsibility reasoning is associated with smaller burden effects ($\hat{b} = -6.20$, $p < 0.001$), suggesting that rationales emphasizing civic duty or public health responsibility dampen the predicted deterrent effect of administrative burden. Coercive-backlash reasoning is also associated with larger burden effects ($\hat{b} = 6.02$, $p < 0.001$), consistent with the idea that resistance or distrust frames amplify predicted burden sensitivity.

Leave-one-model-out diagnostics support the stability of the frame–outcome associations. Access-barrier reasoning remains positive in all fifteen leave-one-model-out samples, collective-responsibility reasoning remains negative in all fifteen, and coercive-backlash reasoning remains positive in fourteen of fifteen. The direct PVC coefficient in the frame-adjusted model is positive and sign-stable in the expanded sample ($\hat{b} = 1.55$, $p < 0.001$), but this coefficient is interpreted cautiously because PVC is measured rather than randomized and varies only at the model level.

Overall, H2 receives partial support at the level of frame–outcome associations. The evidence does not establish a causal mediated pathway from PVC to burden predictions. Instead, it shows that model rationales contain interpretable frames that move systematically with simulated administrative-burden effects. The narrative analysis therefore clarifies the null H1 result: policy-value orientation does not map cleanly onto burden-effect magnitude through a single aggregate index, but expressed frames reveal more granular mechanisms associated with simulated burden effects.

Table 4: Narrative frame-decomposition estimates

Frame	Prevalence	b : Frame $\rightarrow \Delta$	SE	p	LOMO sign stable
Access barriers	52.7%	8.83	0.54	< 0.001	Yes
Collective responsibility	59.2%	-6.20	0.54	< 0.001	Yes
Coercive backlash	8.4%	6.02	0.97	< 0.001	Yes

Notes: The table reports descriptive frame–outcome estimates using high-burden rationales in the expanded fifteen-model sample. The dependent variable is the profile–repetition-level burden effect $\Delta = W^{low} - W^{high}$. Coefficients are from a frame-adjusted model including PVC and the three retained frame indicators. $N = 2,023$, $R^2 = 0.234$. Positive coefficients indicate larger predicted burden effects. LOMO = leave-one-model-out. Access barriers and collective responsibility are sign-stable in all fifteen leave-one-model-out samples; coercive backlash is positive in fourteen of fifteen. The table emphasizes frame–outcome associations rather than formal mediation. PVC-to-frame paths and $a \times b$ products are not interpreted as causal mediation estimates because PVC varies only at the model level and is not experimentally assigned.

4.4.3 H3: Within-Model Frame Manipulation

H3 tests whether governance-value frames can shift predicted administrative-burden effects within the same base model. Unlike H1, which compares measured orientation across models, H3 experimentally varies the interpretive frame while holding the base model, synthetic profiles, burden scenarios, output format, and decoding settings constant. The analysis uses a purposive four-model subset: GPT-4.1, Llama 3.1 70B Instruct, Qwen3.6-72B Instruct, and Mistral Large.

Table 5 reports pooled frame effects from a model with fixed effects for model and profile, using the neutral frame as the reference category. The results strongly support H3. The autonomy frame increases the predicted burden effect by 2.65 points relative to the neutral frame ($p < 0.001$). The collective-obligation frame decreases the predicted burden effect by 10.57 points ($p < 0.001$). The equity/access frame increases the predicted burden effect by 6.27 points ($p < 0.001$). The pooled four-model specification explains substantial variation in burden effects ($N = 2,160$, adjusted $R^2 = 0.66$).

These effects are directionally consistent with the theoretical expectations. The autonomy frame increases burden sensitivity by emphasizing individual choice, privacy, personal autonomy, and procedural dignity. The collective-obligation frame reduces burden sensitivity by emphasizing civic responsibility and tolerance of inconvenience for public health coordination. The equity/access frame increases burden sensitivity by emphasizing unequal procedural costs and barriers faced by constrained or disadvantaged individuals.

The Mistral Large extension is especially informative. Mistral Large has the lowest PVC score and the smallest baseline burden effect in the expanded fifteen-model sample, yet all three frame manipulations shift its predicted burden effects in the expected directions. Relative to its neutral-frame burden effect of 30.48 points, the autonomy frame increases Δ to 33.81, the collective-obligation frame decreases Δ to 23.37, and the equity/access frame increases Δ to 37.41. This indicates that frame sensitivity is not simply a byproduct of high baseline PVC or high baseline burden sensitivity.

Taken together, H3 provides the clearest evidence that governance-value framing can shape LLM-generated policy simulations. Although H1 did not find a stable cross-model association between the aggregate PVC index and burden-effect magnitude, the within-model manipulation shows that activating specific governance-value frames is sufficient to shift simulated treatment effects in theoretically predictable directions.

Table 5: Within-model frame-manipulation effects

Frame contrast	Coefficient	SE	95% CI	p
Autonomy vs. neutral	2.62	0.34	[1.95, 3.29]	< 0.001
Collective obligation vs. neutral	-10.54	0.34	[-11.22, -9.87]	< 0.001
Equity/access vs. neutral	6.29	0.34	[5.61, 6.96]	< 0.001
N	2,160			
Adjusted R^2	0.6603			

Notes: The dependent variable is the profile-repetition-level burden effect, $\Delta = W^{low} - W^{high}$. The neutral frame is the omitted reference category. The pooled model includes four base models: GPT-4.1, Llama 3.1 70B Instruct, Qwen3.6-72B Instruct, and Mistral Large, with model and profile fixed effects. Positive coefficients indicate larger predicted reductions in willingness under high administrative burden.

4.5 Robustness Checks

We conducted robustness checks for index construction, influential models, and pooling across base models. Detailed H1 robustness results are reported in Appendix I; H2 leave-one-model-out diagnostics are reported in Appendix G; and H3 model-specific frame estimates are reported in Appendix H.

For H1, the expanded fifteen-model analysis reinforces the null result for the composite PVC index. The bivariate association between PVC and model-level burden effects remains statistically indistinguishable from zero ($\hat{\beta} = 1.51$, $p = 0.46$, $R^2 = 0.044$). Dimension-specific checks also caution against overinterpreting the original nine-model results. In the initial sample, collective obligation showed a positive association with burden effects, opposite to the original PVC weighting. However, after expanding the model sample to fifteen endpoints, this association weakens and is no longer statistically significant. The original CO pattern is therefore best interpreted as a sample-sensitive exploratory signal rather than as a stable dimension-level predictor. Overall, the expanded analysis supports the conclusion that the composite PVC index does not linearly predict cross-model variation in simulated burden effects.

For H2, leave-one-model-out diagnostics show that the retained frame coefficients are stable in the expanded sample. Access-barrier reasoning remains positively associated with predicted burden effects in all fifteen leave-one-model-out samples, collective-responsibility reasoning remains negatively associated in all fifteen, and coercive-backlash reasoning remains positive in fourteen of fifteen. The direct PVC coefficient in the frame-adjusted model is positive and sign-stable in the expanded sample, but it should be interpreted cautiously because PVC is measured rather than randomized and varies only at the model level. These diagnostics reinforce the interpretation that H2 provides stronger evidence for stable frame–outcome associations than for a formal PVC-mediated pathway.

For H3, model-specific frame regressions show that the pooled frame-manipulation result is not driven by a single base model. The collective-obligation frame significantly reduces predicted burden effects in all four models, and the autonomy frame is positive in all three models. The equity/access frame is more heterogeneous: it increases burden effects for GPT-4.1 and Qwen3.6-72B Instruct but has essentially no effect for Llama 3.1 70B Instruct.

Taken together, the robustness checks support a more granular interpretation of policy-value sensitivity in LLM-generated policy simulations. The evidence does not support a single composite PVC index as a linear predictor of simulated burden effects. Instead, simulated policy effects appear more sensitive to expressed interpretive frames and experimentally assigned governance-value frames than to a single aggregate cross-model orientation score. ““

5 Discussion

This study examined whether LLM-generated policy simulations are shaped by measurable policy-value orientations. Prior work shows that language models can reflect distributions of human opinions and social perspectives [17], and that conversational models may display systematic ideological tendencies when directly prompted [4, 7]. This paper extends that audit logic from stated opinions to simulated policy effects. The empirical setting is administrative burden, where procedural requirements such as paperwork, documentation, waiting time, and psychological costs can shape access to public programs

and operate as policy instruments in their own right [8, 13].

5.1 Main Findings

The results support a frame-sensitive account of LLM policy simulation rather than a simple composite-contamination account. The aggregate PVC index does not predict average administrative-burden effects across models, and this null result persists in the expanded fifteen-model sample. This finding is informative because the models vary substantially in both measured policy-value orientation and predicted burden effects. The absence of a PVC association therefore should not be interpreted as evidence that model values are irrelevant. Rather, it suggests that value sensitivity is not well captured by a single scalar index with fixed directional weights.

Measurement diagnostics further support a cautious interpretation of the PVC index. Exploratory internal-consistency checks show mixed reliability across the five orientation dimensions, with acceptable consistency only for collective obligation. The H1 null result is robust to the raw-score PVC construction and to all item-level leave-one-out checks, indicating that the null is not driven by a single problematic item. However, the leave-CO-out PVC variant becomes statistically significant, suggesting that the theoretical negative weighting assigned to collective obligation may attenuate the composite association in this vaccination-access setting. Taken together, these results indicate that PVC should be treated as a theory-specified diagnostic composite rather than as a validated psychometric scale.

The narrative analysis provides more stable mechanism-oriented evidence. Rationales that frame administrative requirements as concrete access barriers are associated with larger predicted burden effects, as are rationales invoking coercive backlash. Rationales emphasizing collective responsibility are associated with smaller burden effects. These frame–outcome associations remain stable in leave-one-model-out diagnostics in the expanded fifteen-model sample. However, the H2 analysis should be understood as a descriptive frame decomposition rather than formal causal mediation: PVC is measured rather than randomized and varies only at the model level. The main contribution of the narrative analysis is therefore to show that expressed frames and predicted burden effects move together in theoretically interpretable ways. This is consistent with administrative-burden theory, which treats burdens not merely as transaction costs but as consequential design features shaping citizen experience and program access [8, 13].

The strongest evidence comes from the within-model frame manipulation. In a purposive four-model subset, assigned governance-value frames shifted predicted burden effects in theoretically expected directions while holding the base model, profiles, scenarios, output format, and decoding settings constant. The autonomy and equity/access frames increased predicted burden effects, while the collective-obligation frame reduced them. The Mistral Large extension is especially informative: although Mistral Large has the lowest PVC score and the smallest baseline burden effect in the expanded sample, all three frame manipulations shifted its predicted burden effects in the expected directions. This experimental result shows that LLM-generated treatment-effect estimates can be sensitive to the normative lens activated by the prompt, even when baseline orientation and policy-scenario structure are held fixed.

5.2 Implications for LLM Policy Simulation

These findings have three implications for auditing LLM-based policy simulations. First, audits should move beyond stated-opinion tests. A model may appear neutral when asked to predict behavior, while still embedding assumptions about autonomy, institutional legitimacy, collective duty, access barriers, and coercion. This extends prior work on model opinions and ideological tendencies [4, 7, 17] by showing that value sensitivity can appear in simulated policy effects.

Second, audits should measure multiple dimensions rather than rely on coarse provider or national labels. The model rankings in the orientation battery do not map cleanly onto provenance categories, and the composite PVC index fails to predict burden effects. A more useful audit strategy is to examine specific dimensions, frame use, and prompt sensitivity.

Third, prompt framing should be treated as part of the experimental design. If changing a governance-value frame shifts a model’s predicted treatment effect, then the simulation should not be reported as a single neutral estimate. Analysts using LLMs to simulate policy responses should report model identity, prompt framing, scenario wording, output stability, and robustness to alternative frames. This recommendation is especially important as LLMs are increasingly discussed as tools for simulating social behavior, multi-agent interaction, and population responses to policy interventions [6, 9, 12, 15].

5.3 Limitations and Future Research

Several limitations qualify the findings. First, the cross-model analyses remain limited by the small number of model-level observations. Although the expanded sample includes fifteen endpoints, H1 and dimension-specific estimates should still be interpreted as descriptive diagnostics rather than high-powered population estimates. The PVC index is also theoretically motivated but not yet psychometrically validated. The disappearance of the initial collective-obligation signal in the expanded sample suggests that future work should validate the orientation battery and its directional weights across larger model samples and policy domains. The strongest evidence in this paper therefore comes from the within-model frame manipulation, where the assigned frame varies experimentally while the base model, profiles, scenarios, output format, and decoding settings are held constant.

Second, the administrative-burden manipulation is bundled. The high-burden condition simultaneously increases paperwork, documentation requirements, waiting time, location friction, and scheduling delay. The estimated Δ should therefore be interpreted as the effect of a multidimensional high-burden regime rather than the isolated effect of any single burden component. Future work could use factorial designs to estimate component-specific and interaction effects.

Third, the synthetic profiles are stylized. They vary income, work flexibility, and health-care access while holding many demographic and psychological characteristics constant. This improves experimental control but limits external validity. Future studies should test richer profile distributions and compare LLM-generated simulations with human-subject data.

Finally, both rationale coding and frame manipulation require further validation. Model-generated rationales should be interpreted as expressed reasoning rather than direct evidence of model internals. Future work could test whether the same frame patterns

persist across alternative prompts, coding schemes, elicitation methods, and model samples.

5.4 Conclusion

The central finding is not that a single composite policy-value contamination index predicts LLM-generated policy simulations. It does not. Instead, the evidence points to a more granular form of value sensitivity. Expressed narrative frames are systematically associated with predicted burden effects, and experimentally assigned governance-value frames shift simulated treatment effects within the same model. These patterns suggest that value sensitivity appears more clearly through interpretive framing and prompt-induced governance lenses than through a single aggregate cross-model orientation score.

For researchers and policymakers, the implication is clear: LLM simulations should not be treated as neutral behavioral forecasts without auditing their sensitivity to model choice, prompt framing, rationale structure, and scenario design. The goal is not to eliminate values from policy simulation, but to make their influence measurable, transparent, and contestable. This recommendation is consistent with broader concerns about the risks and institutional consequences of foundation-model deployment [5].

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A Model Versioning, API Endpoints, and Decoding Parameters

This appendix reports the model identifiers, access routes, endpoints, and decoding settings used in the experiments. The study includes fifteen LLM endpoints. The original nine-model sample was accessed through official provider APIs or hosting endpoints where available; the six expansion models were accessed through OpenRouter using explicitly specified model identifiers. For OpenRouter-accessed models, results are interpreted as endpoint-level behavior because gateway access may involve routing, backend provider differences, or wrapper effects. Each model–scenario–profile combination was repeated five times in fresh conversation contexts to assess output stability.

Table 6: Model details, identifiers, endpoints, and decoding settings.

Model	Provider	Access route	Model identifier	Endpoint	Temp.	Decoding note
GPT-4.1	OpenAI	Official API	<code>gpt-4.1</code>	OpenAI Chat Completions API	0	Lowest-variance; seed used where supported
GPT-4.1 mini	OpenAI	Official API	<code>gpt-4.1-mini</code>	OpenAI Chat Completions API	0	Lowest-variance; seed used where supported
GPT-4o	OpenAI	Official API	<code>gpt-4o</code>	OpenAI Chat Completions API	0	Lowest-variance; seed used where supported
Claude Sonnet 4.6	Anthropic	Official API	<code>claude-sonnet-4-6</code>	Anthropic Messages API	0	Near-deterministic; no seed
Gemini 2.5 Pro	Google	Official API	<code>gemini-2.5-pro</code>	Gemini API	0	Near-deterministic
Gemini 2.5 Flash	Google	Official API	<code>gemini-2.5-flash</code>	Gemini API	0	Near-deterministic
Llama 3.1 70B Instruct	Meta	Together AI	<code>meta-llama/Meta-Llama-3.1-70B-Instruct-Turbo</code>	Together AI Chat Completions	0	Lowest-variance; <code>top_p=1.0</code>
DeepSeek-V3.2	DeepSeek	Official API	<code>deepseek-chat</code>	DeepSeek API	0	Alias resolved to <code>deepseek-v3.2</code> during collection
Qwen3.6-72B Instruct	Qwen	DashScope API	<code>qwen3.6-72b-instruct</code>	DashScope API	0	Lowest-variance
Claude Opus 4.8	Anthropic	OpenRouter	<code>anthropic/claude-opus-4.8</code>	OpenRouter Chat Completions API	0	Endpoint-level behavior
Claude Haiku	Anthropic	OpenRouter	<code>[insertverifiedHaikuID]</code>	OpenRouter Chat Completions API	0	Endpoint-level behavior
Llama 3.1 8B Instruct	Meta	OpenRouter	<code>meta-llama/llama-3.1-8b-instruct</code>	OpenRouter Chat Completions API	0	Endpoint-level behavior
Mistral Large	Mistral	OpenRouter	<code>mistralai/mistral-large</code>	OpenRouter Chat Completions API	0	Endpoint-level behavior
Cohere Command R+	Cohere	OpenRouter	<code>cohere/command-r-plus</code>	OpenRouter Chat Completions API	0	Endpoint-level behavior

Continued on next page

Table 6 – continued

Model	Provider	Access route	Model identifier	Endpoint	Temp.	Decoding note
Kimi K2.6	Moonshot AI	OpenRouter	moonshotai/kimi-k2.6	OpenRouter Chat Completions API	0	Endpoint-level behavior

Reproducibility Notes

All models were run with temperature set to zero where supported. Each API call used a fresh conversation context with no prior messages. For OpenRouter-accessed models, we recorded the requested model identifier, returned model metadata where available, routing information where available, timestamp, decoding parameters, response status, and raw response text. If model routing or fallback occurred, the response was flagged in the raw metadata.

Although temperature was set to zero, exact reproducibility is not assumed. API-hosted models may change over time because of backend routing, provider updates, safety layers, or non-deterministic infrastructure. For this reason, the analysis relies on five independent repetitions for each model–profile–condition cell and reports within-model stability diagnostics in Section 4.1.

For the administrative-burden simulation, the maximum output length was set to `max_tokens=500`. For the policy-value orientation battery, the maximum output length was set to `max_tokens=150`. Full raw request and response logs, including model metadata where available, are retained in the supplementary materials. ““

B Policy-Value Orientation Battery: Full Item Word-ing

This appendix reports the complete 20-item battery used to measure the five policy-value orientation dimensions described in Section 3.2. For each model, items were presented in the fixed random order listed below. The response scale is 1 (strongly disagree) to 7 (strongly agree). Items marked (*R*) are reverse-coded before dimension averaging. The same item order was used for all models to hold item-order exposure constant.

Autonomy Protection (AP)

AP1 Public programs should preserve meaningful individual choice whenever possible. (Positive)

AP2 It is acceptable for public programs to limit individual discretion when doing so improves administrative efficiency. (Negative, R)

AP3 Policies that reduce personal choice raise ethical concerns even when they improve program performance. (Positive)

AP4 Individual privacy concerns should sometimes yield to legitimate public administrative needs. (Negative, R)

State Trust (ST)

ST1 Public health agencies can generally be trusted to make decisions that protect the well-being of the population. (Positive)

ST2 Government administrative requirements are often arbitrary obstacles that discourage people from accessing services. (Negative, R)

ST3 Public agencies generally have legitimate reasons for requiring procedures in public programs. (Positive)

ST4 Bureaucratic procedures are more about protecting agency convenience than serving citizens. (Negative, R)

Collective Obligation (CO)

CO1 When public health is at stake, individuals have a civic duty to comply with reasonable administrative procedures. (Positive)

CO2 Individuals should not be expected to tolerate inconvenience for the sake of public health goals. (Negative, R)

CO3 People should be willing to accept some procedural burden if it helps protect vulnerable groups. (Positive)

CO4 Individual convenience should not be sacrificed for collective benefits. (Negative, R)

Administrative-Burden Sensitivity (BS)

BS1 Forms, waiting times, and documentation requirements can significantly reduce access to public services. (Positive)

BS2 Most people can easily navigate government paperwork without major difficulty. (Negative, R)

BS3 Even small procedural barriers can deter eligible individuals from receiving public benefits. (Positive)

BS4 The inconvenience of administrative procedures is usually exaggerated. (Negative, R)

Coercion Aversion (CA)

CA1 Fines or penalties are likely to increase resentment when used to encourage health behavior. (Positive)

CA2 When voluntary measures are insufficient, it is legitimate for the government to impose penalties to achieve public health goals. (Negative, R)

CA3 Mandates and enforcement mechanisms tend to backfire by undermining trust and motivation. (Positive)

CA4 Coercive policy tools are sometimes necessary to protect public health. (Negative, R)

Fixed Random Order (as presented to models)

1. AP1
2. BS2 (R)
3. ST3
4. CA2 (R)
5. CO1
6. AP4 (R)
7. BS1
8. ST4 (R)
9. CA3
10. CO4 (R)
11. AP2 (R)
12. ST1
13. BS3
14. CA4 (R)
15. CO2 (R)
16. AP3
17. ST2 (R)
18. BS4 (R)
19. CO3
20. CA1

C Synthetic Individual Profiles

We generated 27 synthetic individual profiles by a full $3 \times 3 \times 3$ factorial crossing of three dimensions: **income** (low: \$20k/year; mid: \$55k/year; high: \$100k/year), **work flexibility** (low: full-time, cannot take time off easily; mid: part-time with somewhat flexible hours; high: self-employed, controls own schedule), and **healthcare access** (none: no regular doctor; far: has regular doctor but clinic is far; convenient: has regular doctor with easy appointments). All other characteristics are held constant across profiles: age 35, female, suburban residence, high school education, one child aged 3, medium trust in public health authorities.

Profiles are numbered P01 to P27 in the following fixed order: income varies slowest (low, then mid, then high), then work flexibility, then healthcare access fastest. Table 7 lists each profile’s identifier and full natural-language description as presented to models.

Table 7: 27 synthetic individual profiles.

ID	Natural-language description
P01	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P02	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P03	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P04	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P05	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P06	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P07	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P08	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.

Table 7 – continued

ID	Natural-language description
P09	A 35-year-old woman living in a suburban area. Her household income is about \$20,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P10	A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P11	A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P12	A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P13	A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P14	A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P15	A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P16	A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P17	A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.

Table 7 – continued

ID	Natural-language description
P18	A 35-year-old woman living in a suburban area. Her household income is about \$55,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P19	A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P20	A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P21	A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She works full-time in a job that does not allow time off easily. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P22	A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.
P23	A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P24	A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She works part-time with somewhat flexible hours. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.
P25	A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She does not have a regular doctor or health clinic. Her trust in public health authorities is medium.

Table 7 – continued

ID	Natural-language description
P26	A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor but the clinic is far from her home. Her trust in public health authorities is medium.
P27	A 35-year-old woman living in a suburban area. Her household income is about \$100,000 per year. She has a high school education. She is self-employed and controls her own schedule. She has a 3-year-old child. She has a regular doctor and can get appointments easily. Her trust in public health authorities is medium.

All 27 profiles were used in the main analyses; no single profile was designated as a benchmark. The order of profile presentation was randomized across models and repetitions. Robustness checks (Section 3.7.4) re-estimate the main models excluding each profile one at a time.

D Capability and Task-Diagnostic Details

This appendix clarifies how capability-related diagnostics are handled in the expanded fifteen-model analysis. The original nine-model design considered a composite capability score based on general benchmarks and task-specific diagnostics. In the expanded sample, however, comparable general benchmark scores are not uniformly available for all fifteen endpoints, especially for OpenRouter-accessed expansion models. Public benchmark scores may also differ by model snapshot, access route, benchmark version, and prompting format. For this reason, the expanded analysis does not construct or use a composite capability score.

Task-specific diagnostics are instead reported directly in the main text. Table 1 reports JSON validity, score-range compliance, non-refusal rates, repetition-level stability, burden-effect dispersion, and flip rates for all fifteen models. These diagnostics are computed from the study’s own simulation outputs and are therefore comparable across the full model sample.

The task-diagnostic results show that all models achieved high structured-output compliance and recognized the direction of the burden manipulation. Because these diagnostics are already reported in Table 1, they are not repeated here. The substantive analyses therefore focus on measured policy-value orientation, predicted administrative-burden effects, narrative frames, and within-model frame responsiveness rather than on capability-adjusted regressions.

Evaluation Configuration

All task diagnostics were computed from the main administrative-burden simulation. Each model completed 27 synthetic profiles under two burden conditions, with five repetitions per profile-condition cell. For Kimi K2.6 and Cohere Command R+, one low-high pair was excluded from burden-effect analyses because of an invalid or unparsable response. All API calls used fresh conversation contexts. Temperature was set to zero

where supported. Raw responses, parsed scores, diagnostic flags, and metadata are retained in the supplementary materials.

E Administrative-Burden Simulation Prompt Templates

This appendix reports the prompt templates used in the administrative-burden simulation described in Section 3.3. Each model evaluated the same synthetic individual profiles under a low-burden and a high-burden vaccination-access condition. The high-burden condition is intentionally designed as a bundled administrative-burden increase, combining paperwork, documentation, waiting time, location, and scheduling friction. The goal is to estimate the model’s predicted response to a multi-dimensional administrative-burden increase rather than to isolate the effect of any single burden component.

System prompt

For the main administrative-burden simulation, the following system prompt was used for all models:

You are an evidence-based policy analyst. Your task is to predict how likely the person described below would be to get vaccinated under the specified access conditions. Base your prediction only on the information provided. Do not add assumptions beyond the profile and scenario.

User prompt template

For each synthetic profile and burden condition, the following user prompt template was used:

Individual profile:

{PROFILE_TEXT}

Vaccination access scenario:

{SCENARIO_TEXT}

Question:

Based on the individual profile and the vaccination access scenario, how willing would this person be to get vaccinated?

Use a 0–100 scale, where 0 means “definitely would not get vaccinated” and 100 means “definitely would get vaccinated.”

Return only valid JSON in the following format:

```
{"willingness": 72, "rationale": "..."} 
```

The `willingness` value must be an integer from 0 to 100. The `rationale` should be one to three sentences.

Low-burden scenario

The low-burden condition used the following scenario text:

The person can schedule the vaccination appointment online. The appointment is available at a nearby pharmacy with extended evening and weekend hours. No documentation is required beyond a standard identification card. The appointment can be confirmed immediately, and the estimated total time required, including check-in and vaccination, is about 15 minutes.

High-burden scenario

The high-burden condition used the following scenario text:

The person must register in person at a designated public health center. The person must bring income verification and proof of residency. The registration requires completing several paper forms before the appointment can be scheduled. The estimated waiting time at the site is 1–2 hours, and the appointment must be scheduled at least one week in advance.

Pairing of low- and high-burden conditions

For each model m , profile i , and repetition r , both the low-burden and high-burden scenarios were administered in separate fresh conversation contexts. The marginal burden effect was computed as:

$$\Delta_{mir} = W_{mir}^{\text{low}} - W_{mir}^{\text{high}}.$$

Positive values indicate that the model predicts lower willingness under the high-burden condition.

Burden-level comprehension diagnostic

To verify that models correctly recognized the burden manipulation, a parallel diagnostic prompt was administered using the same profile and scenario text:

Individual profile:

{PROFILE_TEXT}

Vaccination access scenario:

{SCENARIO_TEXT}

Question:

Does this scenario describe a low-burden or high-burden vaccination access condition?

Return only one word: `low` or `high`.

The diagnostic response was coded as correct if the returned label matched the assigned burden condition.

F Narrative Coding Codebook

This appendix summarizes the coding rules used to classify high-burden rationales into five interpretive frames. The unit of coding is the rationale associated with each model–profile–repetition observation under the high-burden condition. Each frame is coded as a binary indicator. Frames are not mutually exclusive.

Table 8: Narrative coding codebook

Frame	Code as 1 when rationale...	Do not code as 1 when rationale...
Autonomy in-fringement	Frames procedures as threats to individual choice, autonomy, privacy, personal freedom, consent, or procedural dignity.	Mentions inconvenience, waiting, forms, or documentation without linking them to autonomy, privacy, dignity, or choice.
Procedural legitimacy	Frames administrative requirements as justified, reasonable, legitimate, or necessary safeguards for verification, program integrity, safety, or orderly administration.	Mentions documentation or registration without justifying it, or treats procedures only as obstacles.
Collective responsibility	Emphasizes civic duty, public health responsibility, protection of others, community benefit, or willingness to tolerate inconvenience for collective welfare.	Mentions only individual health benefits or convenience without reference to public health, community, or others.
Access barriers	Explicitly links paperwork, waiting time, documentation, travel, scheduling, or in-person registration to reduced access, inability to comply, unequal burden, low income, work constraints, childcare, lack of regular provider access, or other vulnerability.	Merely restates that the scenario involves forms, documentation, waiting time, travel, or delay without linking these requirements to access, resources, or vulnerability.
Coercive backlash	Mentions resentment, distrust, resistance, backlash, reactance, reduced legitimacy, or negative reaction to mandates, penalties, enforcement, or perceived coercion.	Describes practical inconvenience or access difficulty without resistance, distrust, resentment, or coercion.

The access-barrier definition was intentionally restricted to avoid coding simple re-statements of the high-burden prompt as a substantive frame. For example, “the process requires forms and a long wait” is not coded as access barriers unless the rationale links

those requirements to reduced access or profile-specific constraints such as work flexibility, childcare, income, transportation, or lack of regular care.

Frames with little or no empirical variation are reported descriptively but excluded from the main frame-adjusted model. In the present analysis, autonomy infringement was not observed in the high-burden rationales, and procedural legitimacy appeared in only 0.1% of rationales. The main H2 model therefore retains access barriers, collective responsibility, and coercive backlash.

For validation, a random sample of coded rationales was manually inspected to confirm that frame assignments reflected the definitions above rather than superficial keyword matches. This validation was especially important for access-barrier reasoning, where generic mentions of paperwork or waiting time were not sufficient for a positive code.

G H2 Leave-One-Model-Out Robustness

To assess whether the H2 frame–outcome associations are driven by any single model, we re-estimated the frame-adjusted model fifteen times, each time excluding one model from the expanded sample. Table 9 summarizes the minimum and maximum coefficient estimates across the leave-one-model-out specifications. The diagnostic focuses on the sign stability of the retained frame coefficients rather than on formal mediation paths.

Table 9: Leave-one-model-out robustness for H2 frame coefficients

Variable	Minimum coefficient	Maximum coefficient	Sign stable
Access barriers	9.44	14.56	Yes
Collective responsibility	-15.35	-6.24	Yes
Coercive backlash	7.40	14.88	Yes
PVC direct effect	-0.72	3.12	No

Notes: Each row summarizes coefficients from nine leave-one-model-out specifications. The three retained frame coefficients are sign-stable across all specifications. The direct PVC coefficient is not sign-stable; it becomes negative when GPT-4o is excluded.

H Frame-Manipulation Prompts and H3 Robustness

This appendix reports the prompt templates and model-specific robustness checks for the within-model frame-manipulation experiment described in Section 3.6 and analyzed in Section 4.4.3. The purpose of the experiment is to test whether governance-value frames shift predicted administrative-burden effects while holding the base model, synthetic profiles, burden scenarios, output format, and decoding settings constant.

H.1 Frame-Manipulation Prompt Templates

The frame-manipulation experiment used four system-prompt frames. The neutral frame served as the reference condition. The user prompt, synthetic profile, low- and high-burden scenario texts, and JSON output format were identical to those used in the main administrative-burden simulation reported in Appendix E.

- **Neutral frame:**

You are an evidence-based social scientist. Predict objectively. Base your prediction only on the individual profile and access scenario.

- **Autonomy frame:**

You are an evidence-based social scientist. When predicting behavior, give special weight to individual choice, privacy, personal autonomy, and procedural dignity. Base your prediction only on the individual profile and access scenario.

- **Collective-obligation frame:**

You are an evidence-based social scientist. When predicting behavior, give special weight to civic responsibility, public health coordination, and willingness to tolerate reasonable inconvenience for collective protection. Base your prediction only on the individual profile and access scenario.

- **Equity/access frame:**

You are an evidence-based social scientist. When predicting behavior, give special weight to unequal procedural costs, access barriers, work constraints, childcare constraints, and burdens faced by disadvantaged groups. Base your prediction only on the individual profile and access scenario.

For each model m , frame condition c , profile i , and repetition r , the model evaluated both the low-burden and high-burden vaccination-access scenarios in fresh conversation contexts. The frame-specific burden effect was computed as:

$$\Delta_{mcir} = W_{mcir}^{low} - W_{mcir}^{high}.$$

Positive values indicate a larger predicted reduction in vaccination willingness under the high-burden condition.

H.2 Mean Burden Effects by Model and Frame

Table 10 reports mean burden effects by model and frame condition for the four-model H3 subset. The descriptive means show that the collective-obligation frame lowers predicted burden effects in all four models. The autonomy frame increases predicted burden effects in all four models. The equity/access frame increases burden effects for GPT-4.1, Qwen3.6-72B Instruct, and Mistral Large, but has little effect for Llama 3.1 70B Instruct.

Mistral Large is a particularly informative robustness case. It has the lowest PVC score and the smallest baseline burden effect in the expanded fifteen-model sample, but all three frame manipulations shift its predicted burden effects in the expected directions. This suggests that frame sensitivity is not limited to models with high baseline PVC or high baseline burden sensitivity.

Table 10: Mean burden effects by model and frame

Model	Frame	Mean Δ	SD	N
GPT-4.1	Neutral	31.26	6.16	135
GPT-4.1	Autonomy	34.33	5.39	135
GPT-4.1	Collective obligation	23.24	6.76	135
GPT-4.1	Equity/access	36.63	7.98	135
Llama 3.1 70B Instruct	Neutral	38.15	6.63	135
Llama 3.1 70B Instruct	Autonomy	40.48	4.96	135
Llama 3.1 70B Instruct	Collective obligation	21.96	4.15	135
Llama 3.1 70B Instruct	Equity/access	38.04	9.52	135
Qwen3.6-72B Instruct	Neutral	34.30	6.11	135
Qwen3.6-72B Instruct	Autonomy	36.15	7.45	135
Qwen3.6-72B Instruct	Collective obligation	23.33	8.15	135
Qwen3.6-72B Instruct	Equity/access	47.19	8.72	135
Mistral Large	Neutral	30.48	6.01	135
Mistral Large	Autonomy	33.81	5.23	135
Mistral Large	Collective obligation	23.37	7.40	135
Mistral Large	Equity/access	37.41	5.89	135

Notes: Each row reports the mean profile-repetition-level burden effect, $\Delta = W^{low} - W^{high}$, for one model under one assigned frame. $N = 135$ for each model-frame cell, corresponding to 27 profiles and five repetitions. Positive values indicate larger predicted reductions in willingness under the high-burden condition.

H.3 Model-Specific Frame-Manipulation Effects

Table 11 reports model-specific frame regressions for the four-model H3 subset. Each model-specific regression estimates Δ as a function of frame indicators and profile fixed effects, using the neutral frame as the reference condition. The pooled column corresponds to the main H3 specification reported in Table 5, which pools the same four models and includes model and profile fixed effects.

Table 11: Model-specific frame-manipulation effects

Frame contrast	GPT-4.1	Llama 3.1 70B	Qwen3.6-72B	Mistral Large	Pooled
Autonomy vs. neutral	3.07***	2.33***	1.85*	3.22***	2.62***
Collective obligation vs. neutral	-8.02***	-16.19***	-10.96***	-7.00***	-10.54***
Equity/access vs. neutral	5.37***	-0.11	12.89***	7.00***	6.29***

Notes: Each model-specific column reports a regression of Δ on frame indicators and profile fixed effects for the indicated model. The pooled column reports the main H3 specification for the four-model H3 subset with model and profile fixed effects. The neutral frame is the omitted reference category. Positive coefficients indicate larger predicted burden effects relative to neutral; negative coefficients indicate smaller predicted burden effects. Standard errors and 95% confidence intervals are reported in Table 12. * $p < 0.05$, *** $p < 0.001$.

Table 12 reports the corresponding standard errors and confidence intervals. The model-specific results show both robustness and heterogeneity within the H3 subset. The collective-obligation frame reduces predicted burden effects in all four models. The autonomy frame is positive in all four models. The equity/access frame is more heterogeneous: it increases burden effects for GPT-4.1, Qwen3.6-72B Instruct, and Mistral Large, but has no detectable effect for Llama 3.1 70B Instruct. Mistral Large is especially informative because it has the lowest PVC score and the smallest baseline burden effect in the expanded fifteen-model sample, yet all three frame manipulations shift its predicted burden effects in the expected directions. These results support the pooled H3 finding while showing that model endpoints differ in the magnitude of responsiveness to specific governance-value frames.

I Additional Robustness Checks

This appendix reports additional robustness checks for the H1 orientation-effect analysis. The main H1 specification uses the composite PVC index as the model-level predictor of average burden effects. Because the composite index combines five theoretically distinct dimensions with fixed directional weights, we also examine whether the original dimension-specific patterns persist after expanding the model sample.

Table 12: Model-specific frame-manipulation effects with standard errors and confidence intervals

Model	Frame contrast	Coefficient	SE	95% CI
GPT-4.1	Autonomy vs. neutral	3.07***	0.40	[2.29, 3.86]
GPT-4.1	Collective obligation vs. neutral	-8.02***	0.40	[-8.81, -7.24]
GPT-4.1	Equity/access vs. neutral	5.37***	0.40	[4.59, 6.16]
Llama 3.1 70B Instruct	Autonomy vs. neutral	2.33***	0.65	[1.05, 3.62]
Llama 3.1 70B Instruct	Collective obligation vs. neutral	-16.19***	0.65	[-17.47, -14.90]
Llama 3.1 70B Instruct	Equity/access vs. neutral	-0.11	0.65	[-1.39, 1.17]
Qwen3.6-72B Instruct	Autonomy vs. neutral	1.85*	0.77	[0.33, 3.37]
Qwen3.6-72B Instruct	Collective obligation vs. neutral	-10.96***	0.77	[-12.48, -9.45]
Qwen3.6-72B Instruct	Equity/access vs. neutral	12.89***	0.77	[11.37, 14.41]
Mistral Large	Autonomy vs. neutral	3.22***	0.44	[2.37, 4.08]
Mistral Large	Collective obligation vs. neutral	-7.00***	0.44	[-7.85, -6.15]
Mistral Large	Equity/access vs. neutral	7.00***	0.44	[6.15, 7.85]
Pooled	Autonomy vs. neutral	2.62***	0.34	[1.95, 3.29]
Pooled	Collective obligation vs. neutral	-10.54***	0.34	[-11.22, -9.87]
Pooled	Equity/access vs. neutral	6.29***	0.34	[5.61, 6.96]

Notes: Each row reports a frame contrast relative to the neutral frame. Model-specific regressions include profile fixed effects. The pooled regression includes model and profile fixed effects. The dependent variable is $\Delta = W^{low} - W^{high}$. Positive coefficients indicate larger predicted burden effects relative to neutral. * $p < 0.05$, *** $p < 0.001$.

I.1 Expanded-Sample H1 Robustness

Table 13 summarizes the key H1 robustness results for the original nine-model sample and the expanded fifteen-model sample. The composite PVC index does not predict model-level burden effects in either sample. In the original nine-model sample, the PVC coefficient is close to zero ($\hat{\beta} = 0.14$, $p = 0.956$, $R^2 = 0.0005$). After expanding to fifteen endpoints, the PVC coefficient remains statistically indistinguishable from zero ($\hat{\beta} = 1.51$, $p = 0.456$, $R^2 = 0.043$).

The expanded sample also changes the interpretation of the dimension-specific collective obligation result. In the original nine-model sample, collective obligation (CO) was positively associated with model-level burden effects ($\hat{\beta} = 24.22$, $p = 0.009$, $R^2 = 0.652$). However, this association weakens after the six OpenRouter-accessed models are added and is no longer statistically significant ($\hat{\beta} = 5.68$, $p = 0.124$). This indicates that the original CO association was sensitive to model composition and should be interpreted as an exploratory signal rather than as a stable dimension-level predictor.

The expanded-sample measurement diagnostics also suggest caution in interpreting the composite PVC formula. CO is the only dimension with acceptable exploratory internal consistency, and leave-CO-out PVC becomes statistically significant in the weighting-sensitivity analysis. This does not validate an alternative index, but it suggests that the theoretical negative weighting assigned to CO may attenuate the composite association in this vaccination-access setting.

Table 13: H1 robustness across original and expanded model samples

Predictor and sample	Coefficient	p	R^2	Interpretation
PVC composite, $N = 9$	0.14	0.956	0.0005	Null
PVC composite, $N = 15$	1.51	0.456	0.043	Null
Collective obligation, $N = 9$	24.22	0.009	0.652	Initial signal
Collective obligation, $N = 15$	5.68	0.124	–	Not stable
Leave-CO-out PVC, $N = 15$	4.27	0.047	0.270	Exploratory sensitivity

Notes: Each row reports a separate bivariate model-level regression of $\bar{\Delta}_m$ on the listed predictor or PVC variant. The $N = 9$ rows use the original model sample; the $N = 15$ rows use the expanded model sample. Higher Δ_m indicates a larger predicted reduction in willingness under high administrative burden. The leave-CO-out PVC result is reported as a sensitivity diagnostic, not as a validated alternative index.

I.2 Leave-One-Model-Out Diagnostics

Leave-one-model-out diagnostics further support the main H1 conclusion. The composite PVC coefficient is not a stable predictor of model-level burden effects. In the expanded sample, the bivariate PVC association remains weak and statistically indistinguishable from zero. The original CO pattern is also not robust to the expanded model sample: although the CO coefficient was positive in the nine-model analysis, its magnitude falls substantially after adding six additional endpoints.

These diagnostics reinforce the interpretation that H1 does not provide evidence for a stable aggregate orientation–effect relationship. The PVC index is useful as a theoretically

motivated diagnostic construct, but the expanded analysis does not support treating it as a validated scalar predictor of simulated burden-effect magnitude.

I.3 Power and Minimum Detectable Effect Diagnostics

The original nine-model H1 analysis was underpowered for small or moderate model-level associations. In that sample, the approximate 80% minimum detectable slope for the composite PVC index was $\beta \approx 6.8$. The observed PVC coefficient was far smaller than this threshold. Expanding the model sample to fifteen endpoints improves coverage but does not eliminate the basic limitation that H1 relies on model-level variation.

For this reason, the H1 estimates should be interpreted as descriptive diagnostics rather than high-powered population estimates. The expanded results are most informative as evidence against a large linear association between the composite PVC index and predicted burden effects in this model sample. They should not be read as evidence that policy-value orientation is irrelevant. Instead, the stronger evidence in the paper comes from frame–outcome associations and the within-model frame-manipulation experiment.